

The seal of the University of Freiburg is a circular emblem. It features a central illustration of a pine tree on a rocky outcrop, with a river flowing in the foreground. The tree is surrounded by a wreath. The text 'UNIVERSITAS FREIBURGENSIS' is inscribed around the top inner edge, and 'FUNDATA 1891' is at the bottom. The words 'FREIHEIT' and 'WEHT' are also visible on the sides.

CS231n: Deep Learning for Computer Vision

Lecture 1: Introduction

Welcome to CS231n



Welcome to CS231n

Course Instructors



Fei-Fei Li Andrej Karpathy

Teaching Assistants



2015

Johnson Yuke Zhu Brett Kuperl Ben Poole

Course Instructors



Justin Johnson

Teaching Assistants

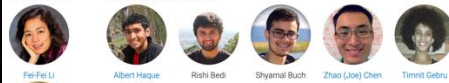


Subhasis Das

2016

Abhinav Gupta

Instructors



Fei-Fei Li

Teaching Assistants



Albert Haque (Head TA)

Justin Johnson

Agam Gupta

De-An Huang

Serena Young

Xingyu Liu

Shayne Longpre

Zelun (Alan) Luo

Lane McIntosh

Oliver Morehead

2017

Amari Poddada

Bogya (Girma) Peng

Ben Poole

Luda Zhao

Instructors



Fei-Fei Li



Rangya Krishna



Danfei Xu



Anshu Bhat

Teaching Assistants



William Shen (Head TA)



Andrew Kondrich



Lynne P. Chappin



Brent Yi



Jonathan Boatz



Fang Yu Lin



Chris Waller



Christina Yuan



Daniel Cai



Damian Mrowca



Rui Wang



Kevin Zakka



Junyoung Seok



Boxiao Pan



Yi Wen



Yiheng Zhang



De-An Huang



Nishant Rai



Karen Yang



Yiheng Zhang

2020

Instructors



Fei-Fei Li



Justin Johnson



Serena Young

Teaching Assistants



Albert Haque (Head TA)



Mark Char



Nishant Rai



Amari Poddada



Chaitanya Asawa



Jim (Linh) Fan



Alexander (Karl) Fu



Mike Roberts



Josh Beal



Alexander (Karl) Fu



Praty Sharma



Fei Xia



Vincent Chen



Pedro Pablo Gonzalez



Ben Zhang



Ben Zhang



Edward Chou



Michelle Guo



Danfei Xu



Ben Zhang



Ben Zhang



John Clow



Jingwei Huang



Ajay Mandekar



Ben Zhang



Ben Zhang

2018

Instructors



Fei-Fei Li



Rangya Krishna



Danfei Xu

Teaching Assistants



Kevin Zakka (Head TA)



Sean Liu



Guangshi Wang



Kevin Zakka (Head TA)



Haofeng Chen



Mandy Lu



Chris Waller



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Rachel Gardner



Nishant Rai



Jessen Zhang



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Samuel Kwong



Geet Sethi



Russell Ma



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Samuel Kwong



Geet Sethi



Russell Ma



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Yichen Li



Lin Shao



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)

2021

Instructors



Fei-Fei Li



Yunchu Li



Ruihan Gao



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)

Teaching Assistants



Ziang Liu (Head TA)



Yuan Gao



Manasi Sharma



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Tannay Agarwal



Yichen Li



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Samir Agarwal



Minjue Huang



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Samuel Clarke



Hao Li



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Zane Durante



Hao Li



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Zane Durante



Hao Li



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)



Kevin Zakka (Head TA)

2023

CS231n's Ten Year Anniversary: Returning Lecturers



Fei-Fei Li
2015-2025



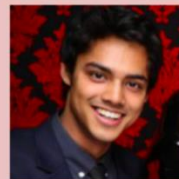
Andrej Karpathy
2015-2016



Justin Johnson
2016-2019, 2025



Serena Yeung
2017-2019



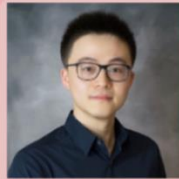
Ranjay Krishna
2020-2021



Danfei Xu
2020-2021



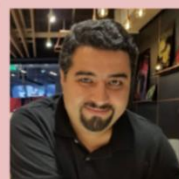
Jiajun Wu
2022



Ruohan Gao
2022-2023



Yunzhu Li
2023

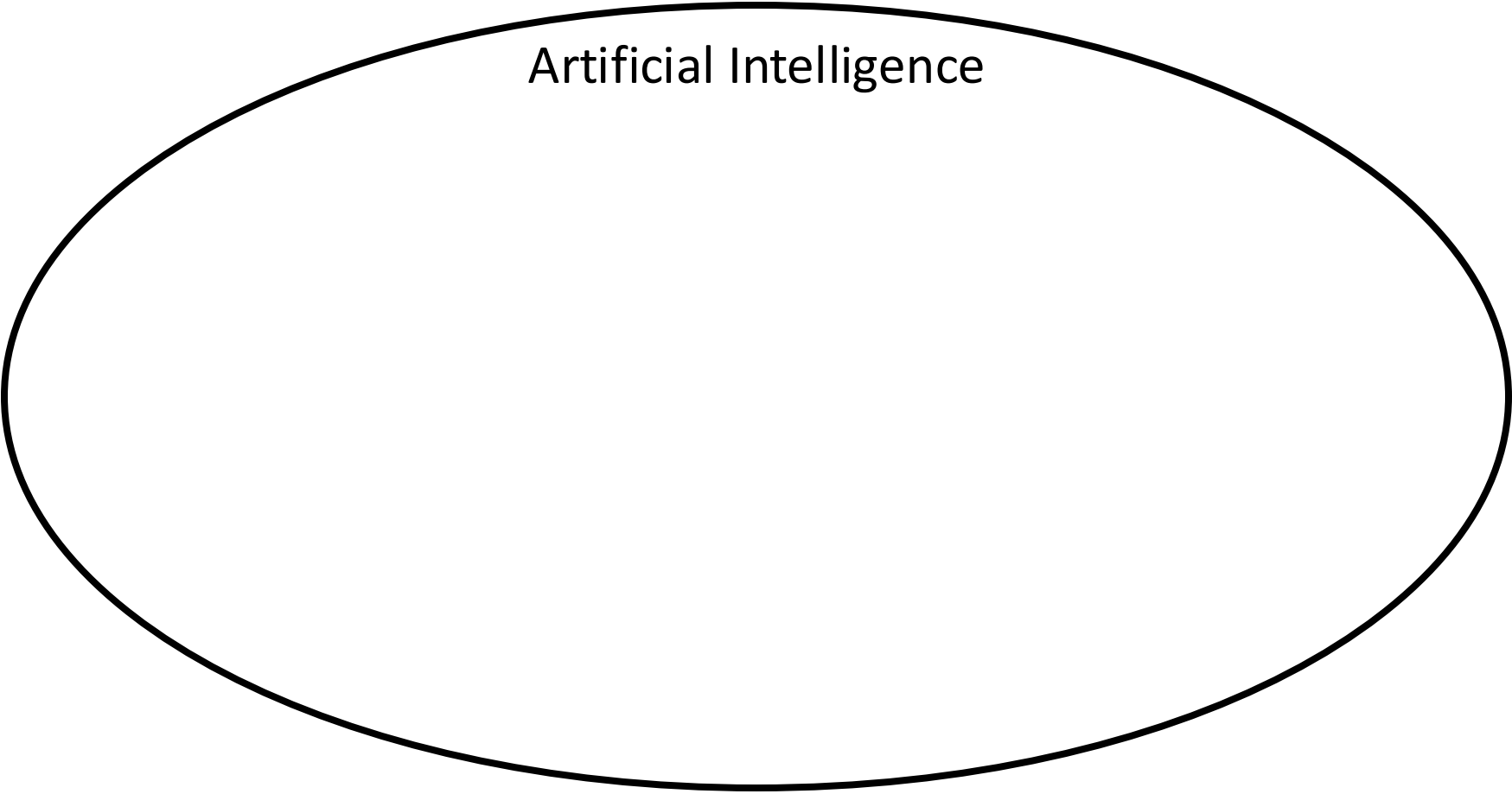


Ehsan Adeli
2024-2025

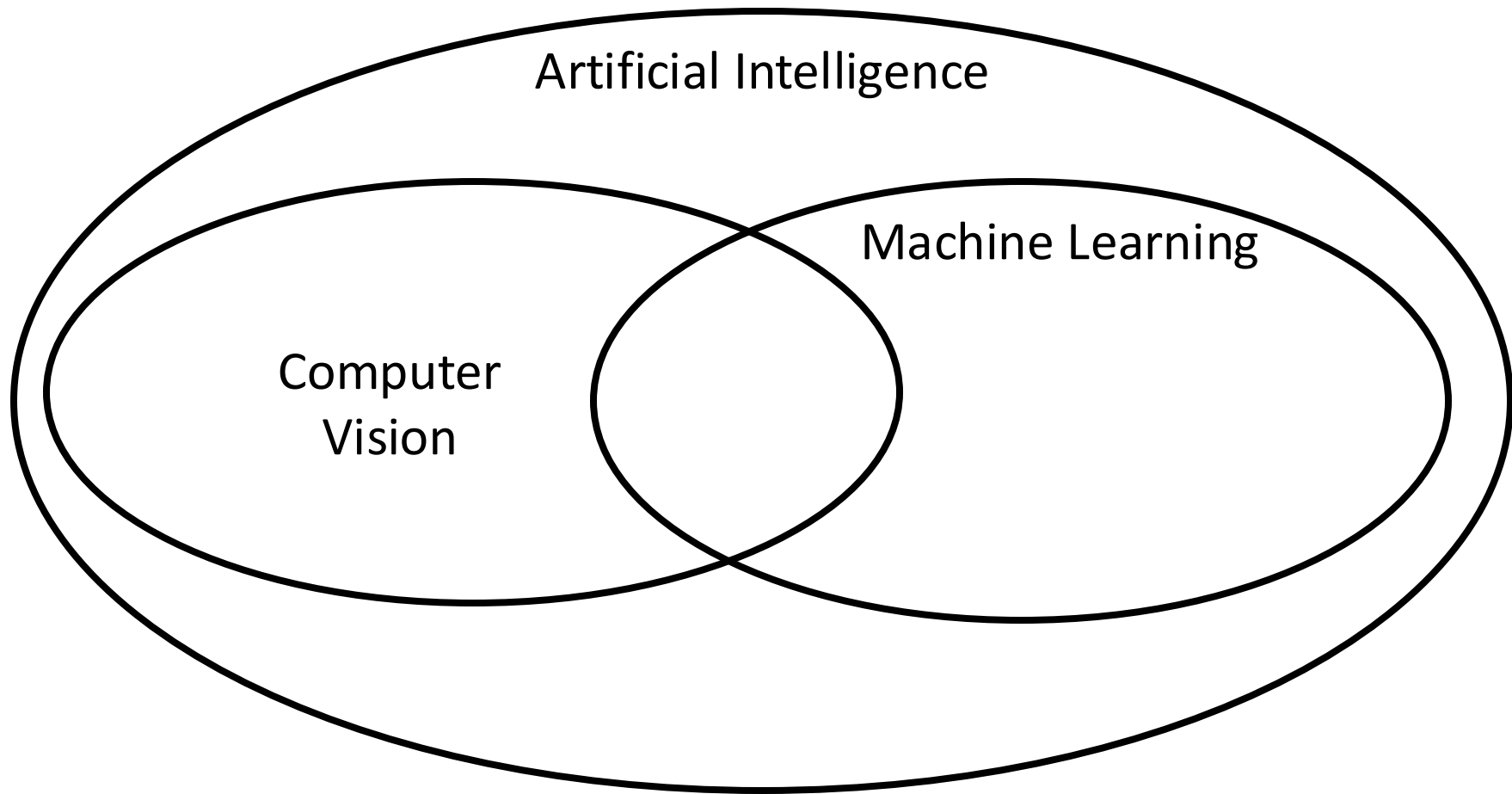


Zane Durante
2024-2025

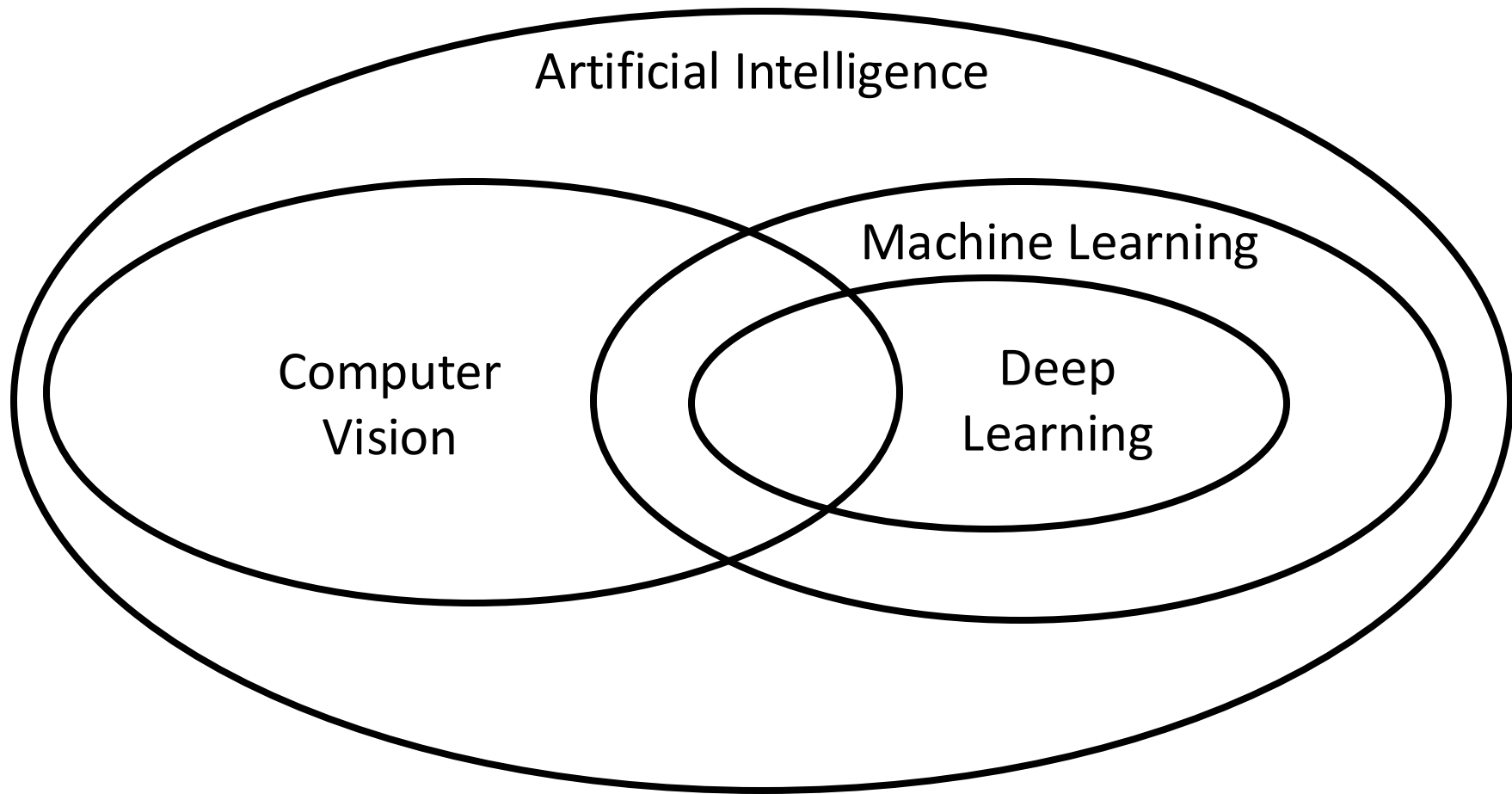
Artificial Intelligence



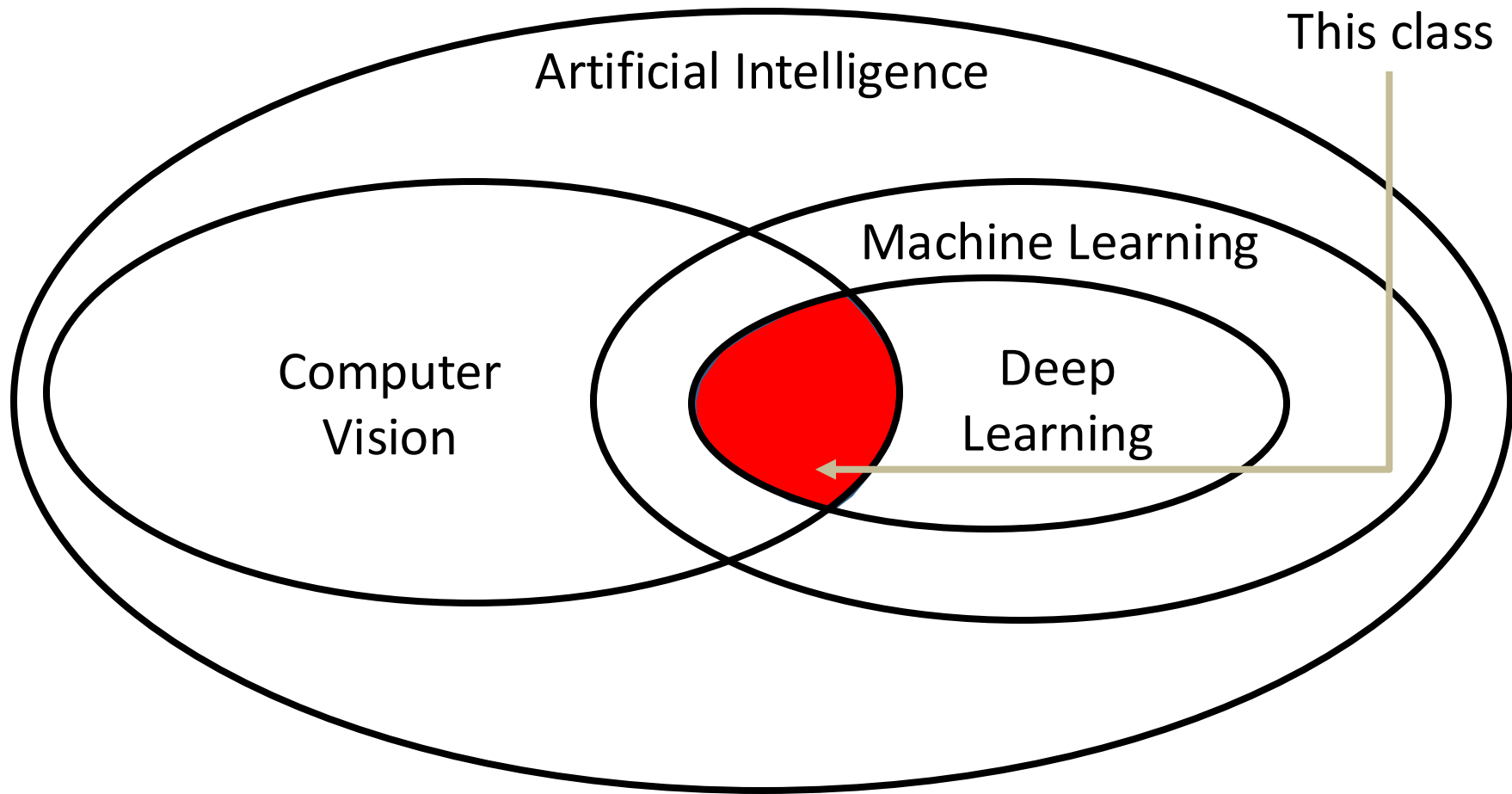
Slide inspiration: Justin Johnson



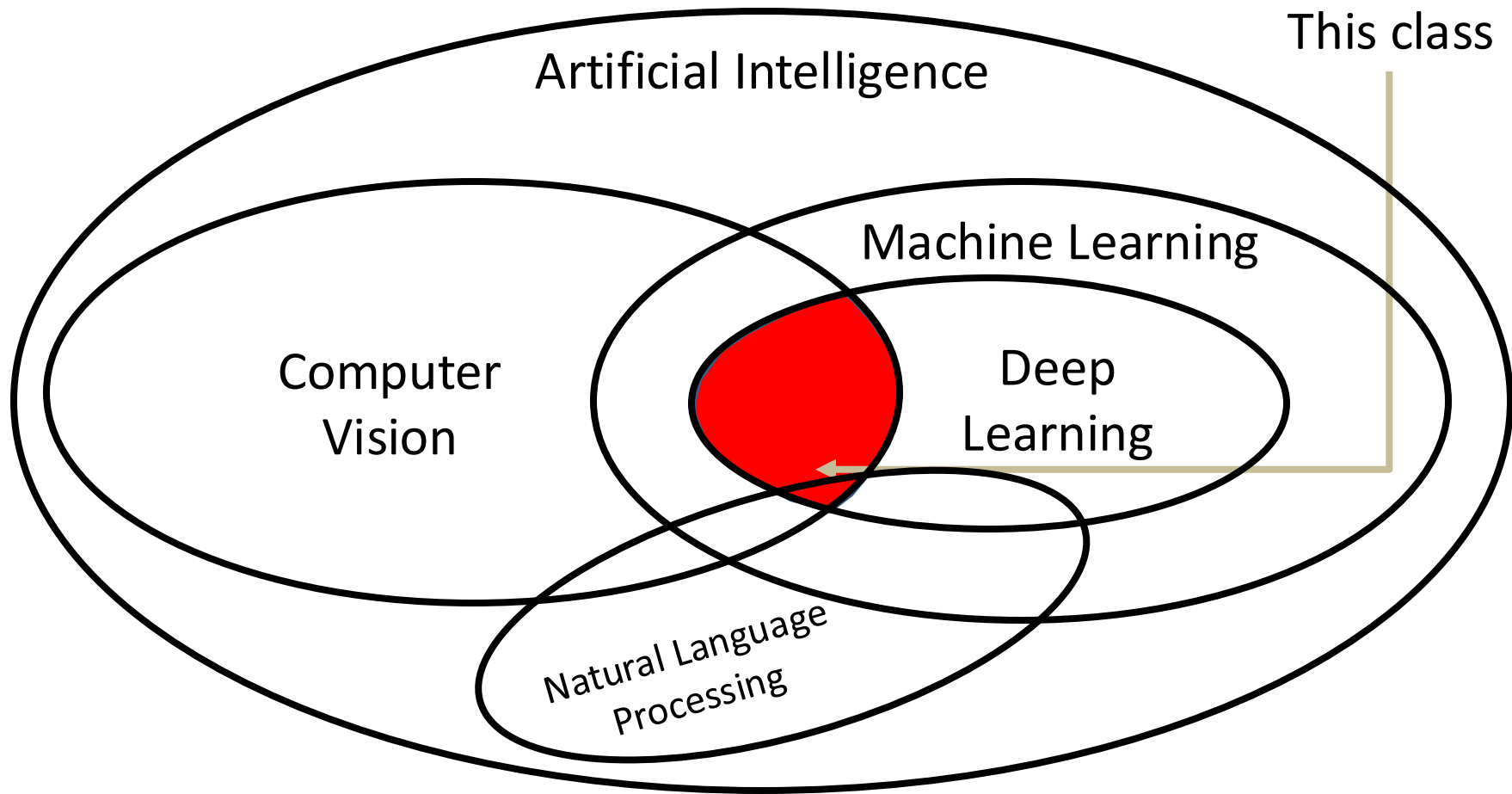
Slide inspiration: Justin Johnson



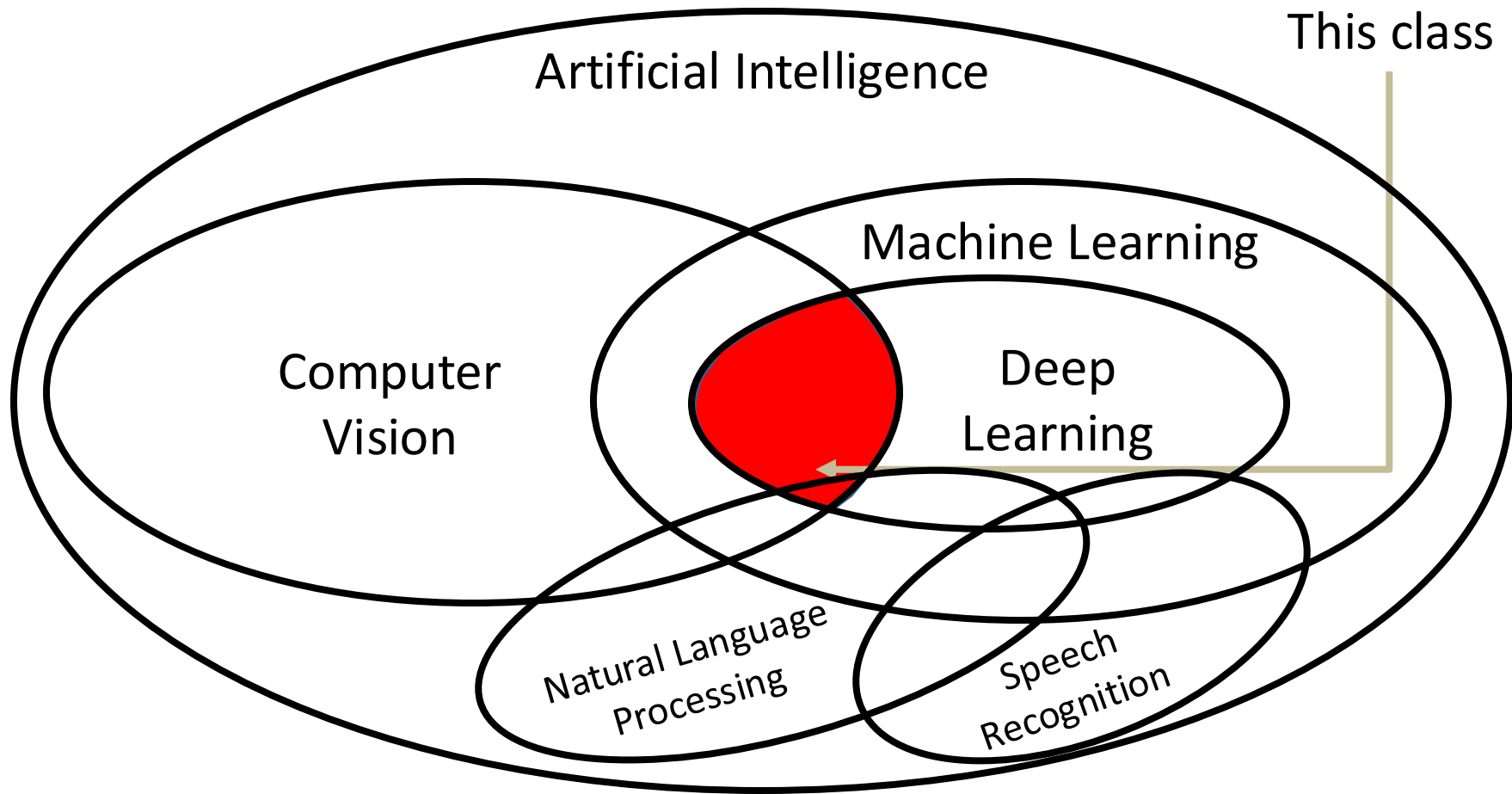
Slide inspiration: Justin Johnson



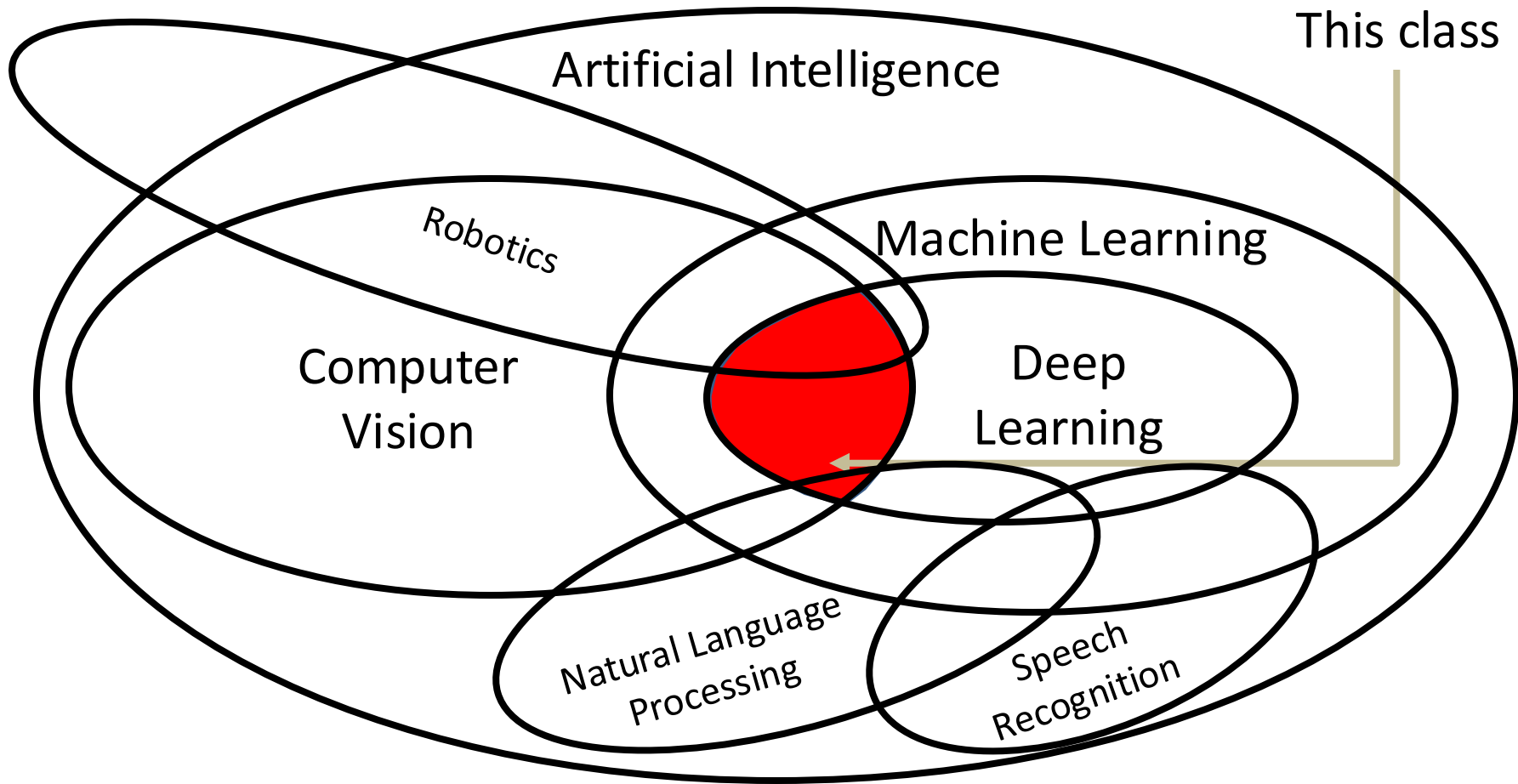
Slide inspiration: Justin Johnson



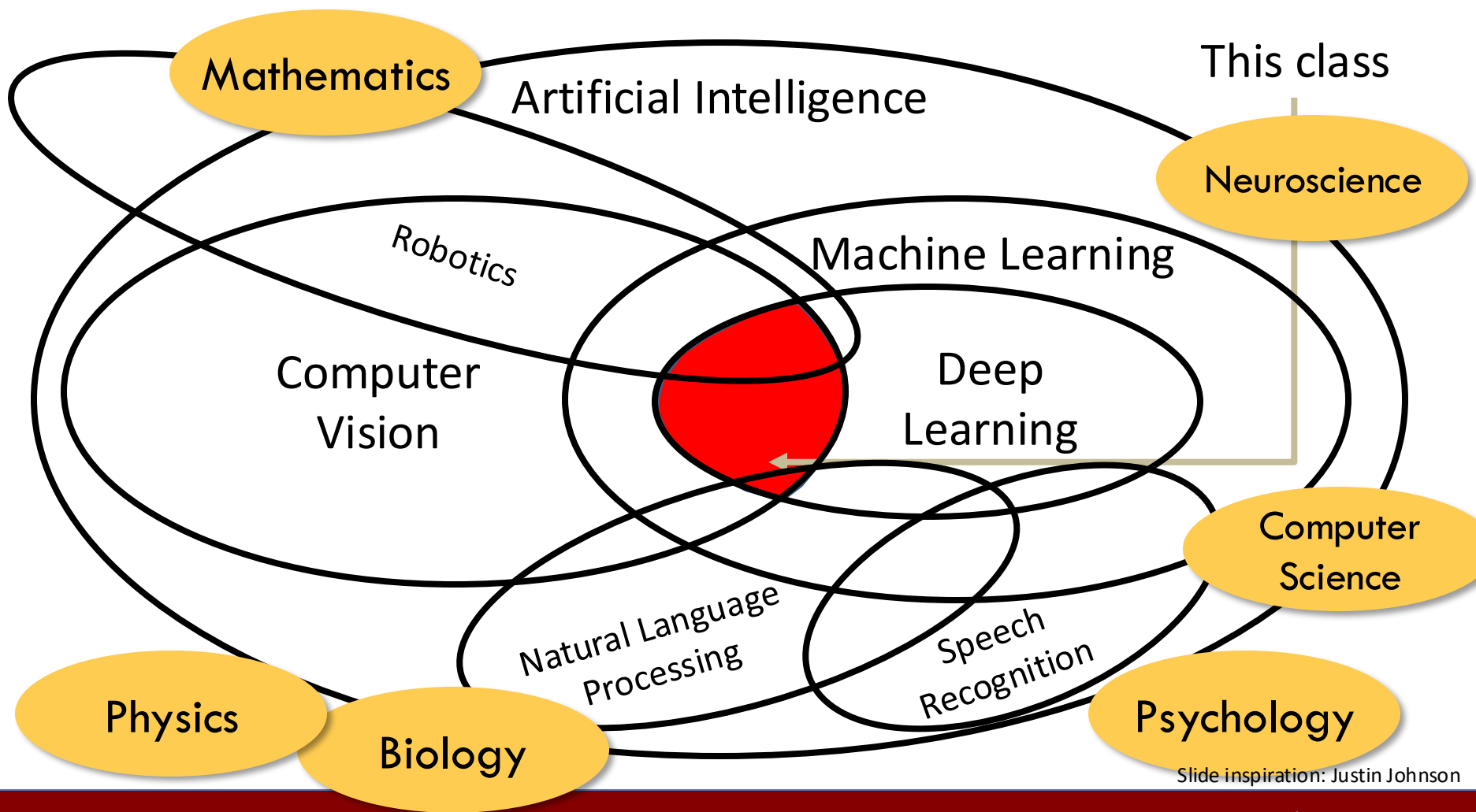
Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson

Today's agenda

- A brief history of computer vision and deep learning
- CS231n overview

Evolution's Big Bang: Cambrian Explosion, 530-540million years, B.C.



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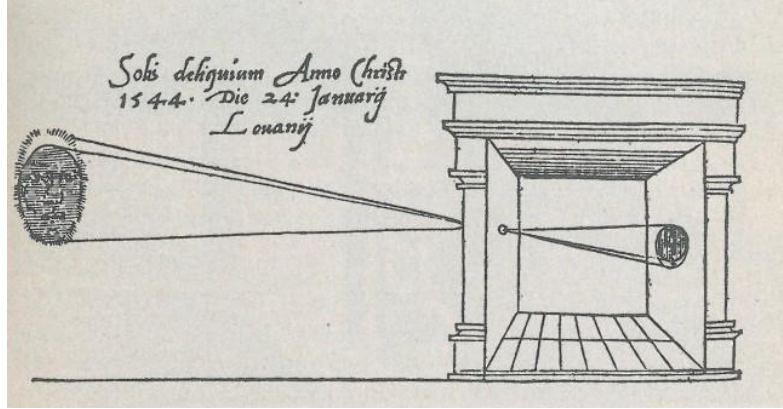


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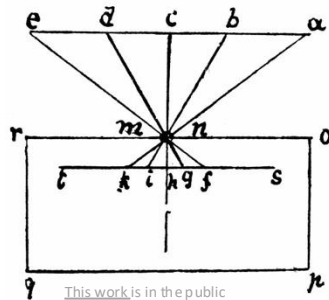


Camera Obscura

Gemma Frisius, 1545



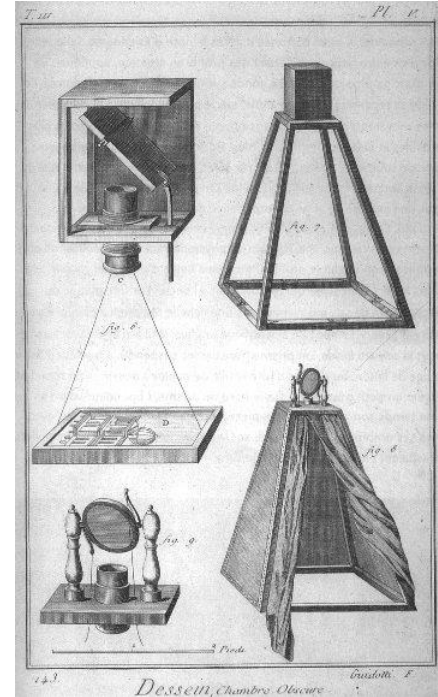
[This work](#) is in the public domain



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Leonardo da Vinci,
16th Century AD

Encyclopedia, 18th Century



Dessein, Chambre Obscure
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Computer Vision is everywhere!



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Where did we come from?

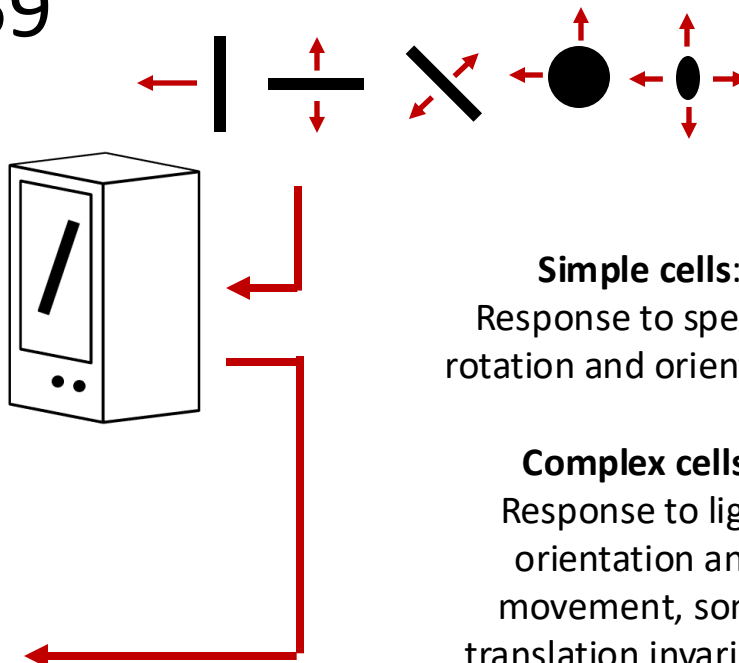
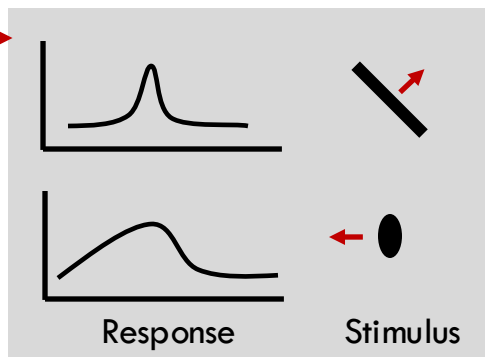
Hubel and Wiesel, 1959

Measure
brain activity



Cat image by CNX OpenStax is licensed under CC BY 4.0; changes made

1959
Hubel & Wiesel

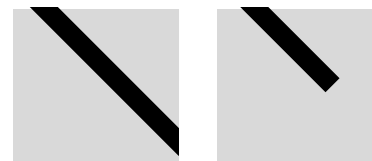


Simple cells:

Response to specific
rotation and orientation

Complex cells:

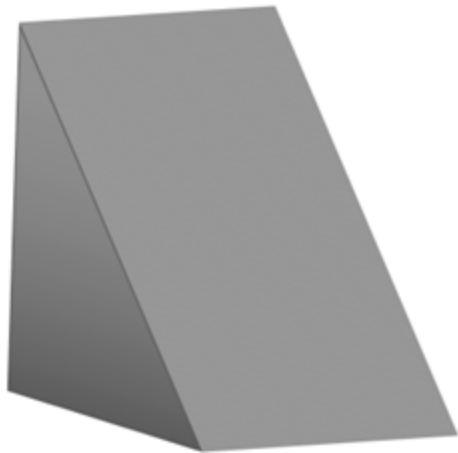
Response to light
orientation and
movement, some
translation invariance



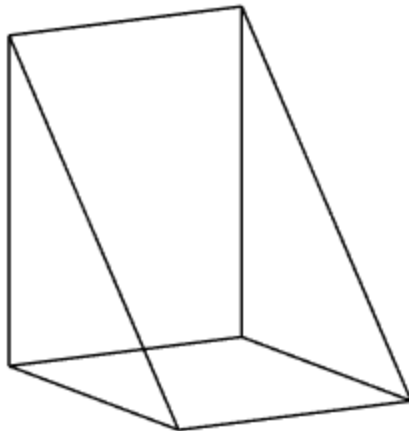
No response

Slide inspiration: Justin Johnson

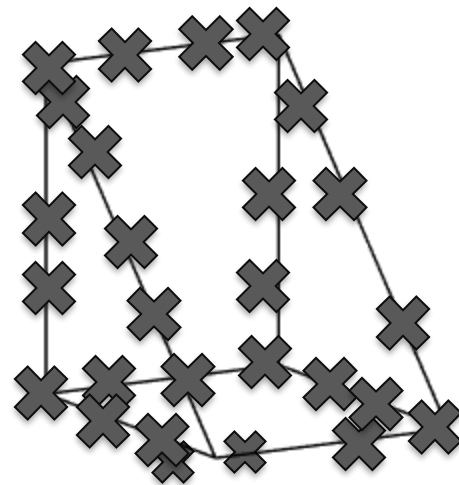
Larry Roberts, 1963



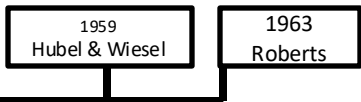
(a) Original picture



(b) Differentiated picture



(c) Feature points selected



Lawrence Gilman Roberts, "Machine Perception of Three-Dimensional Solids", 1963

Slide inspiration: Justin Johnson

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

Seymour Papert

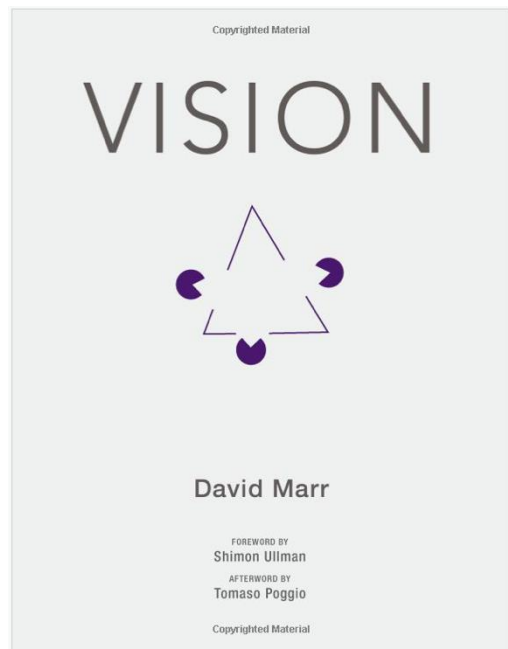
The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

1959
Hubel & Wiesel

1963
Roberts

<https://dspace.mit.edu/handle/1721.1/6125>

Slide inspiration: Justin Johnson

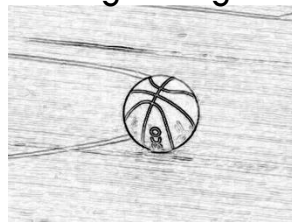


Input image

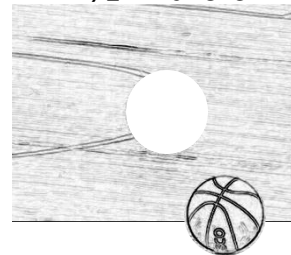


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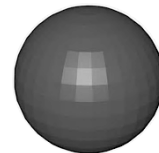
Edge image



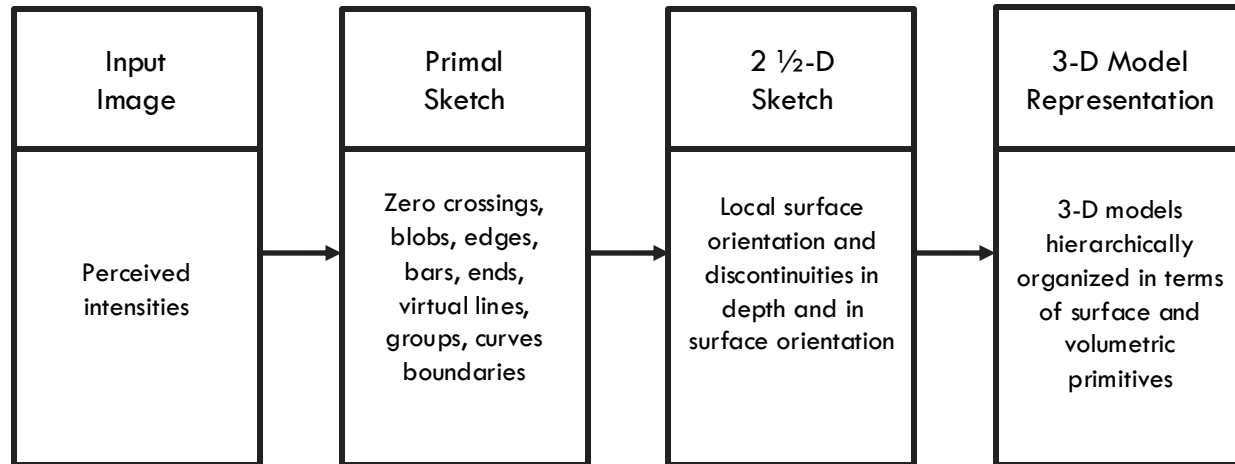
2 1/2-D sketch



3-D model



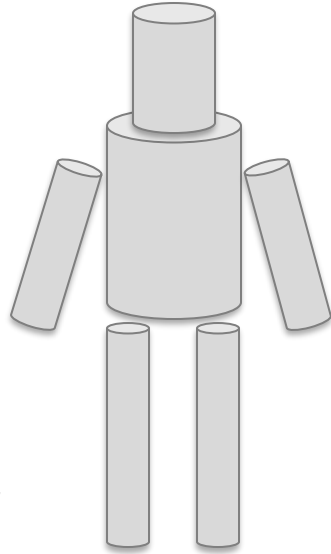
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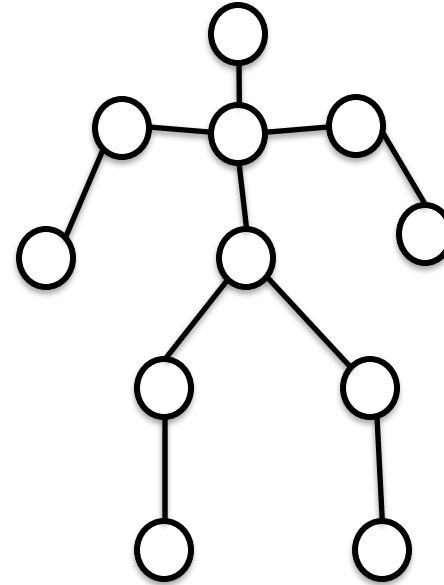
Stages of Visual Representation, David Marr, 1970s

Slide inspiration: Justin Johnson

Recognition via Parts (1970s)



Generalized Cylinders,
Brooks and Binford,
1979

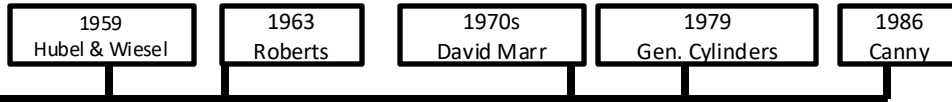
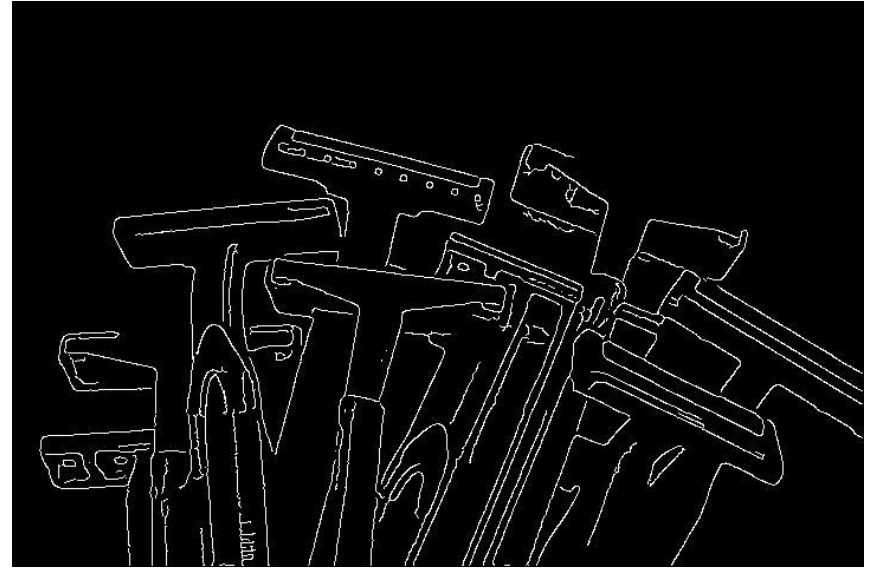


Pictorial Structures,
Fischler and Elshlager, 1973



Slide inspiration: Justin Johnson

Recognition via Edge Detection (1980s)

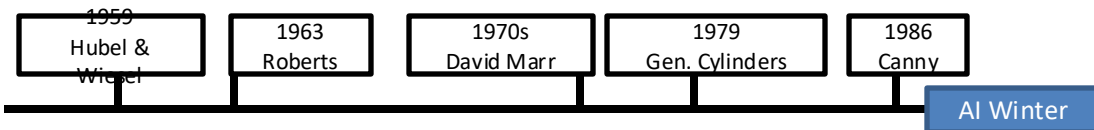


John Canny, 1986
David Lowe, 1987

Slide inspiration: Justin Johnson

Arriving at an “AI winter”

- Enthusiasm (and funding!) for AI research dwindled
- “Expert Systems” failed to deliver on their promises
- But subfields of AI continues to grow
 - Computer vision, NLP, robotics, compbio, etc.



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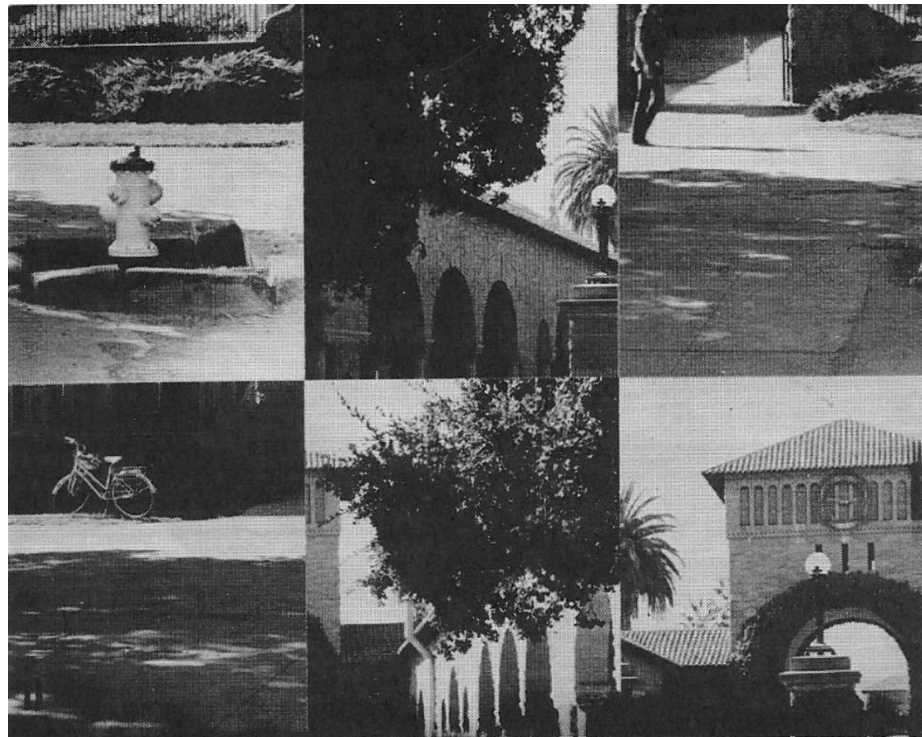
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Slide inspiration: Justin Johnson

In the meantime...seminal work in
cognitive and neuroscience

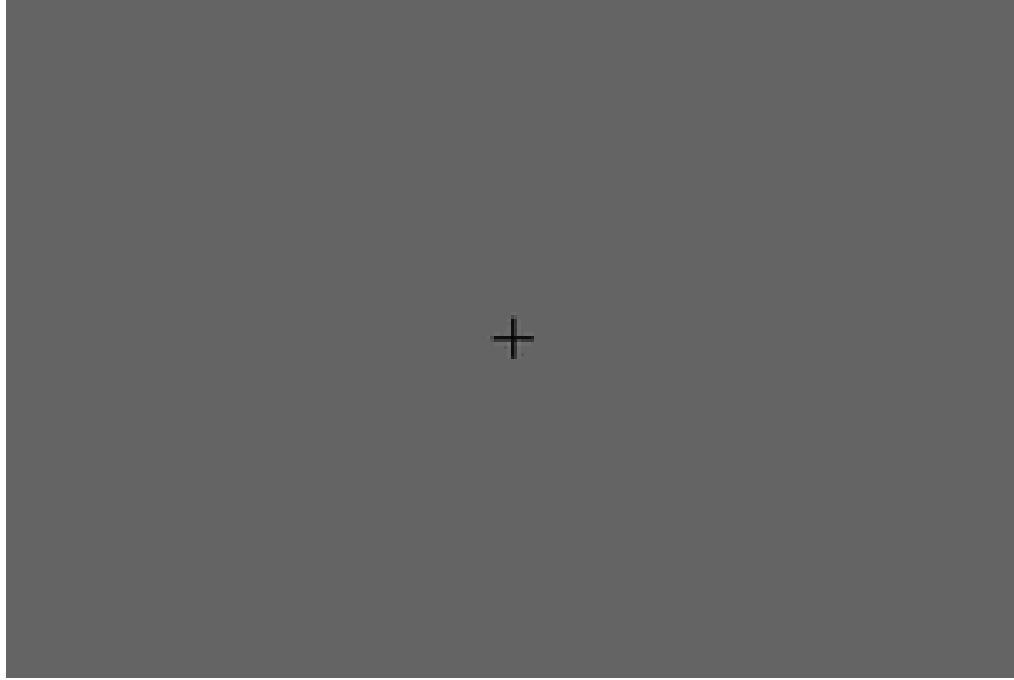
Perceiving Real-World Scenes

Irving Biederman



I. Biederman, *Science*, 1972

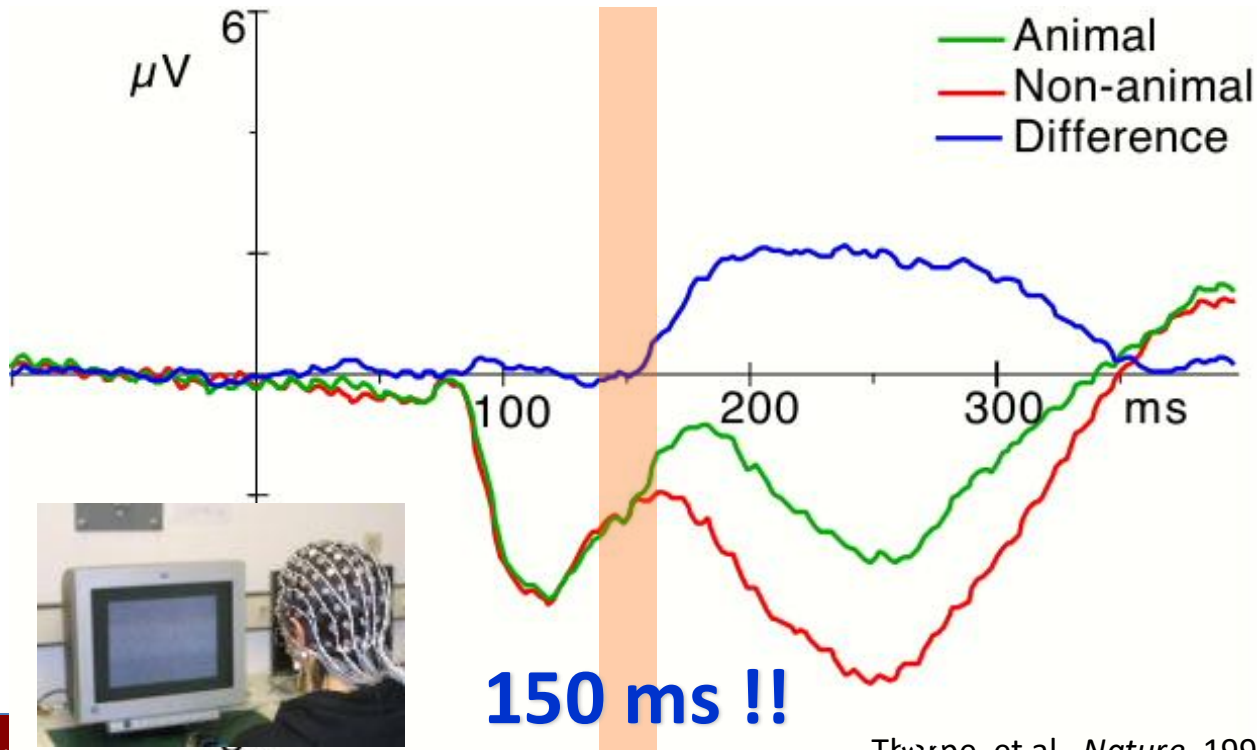
Rapid Serial Visual Perception (RSVP)



Potter, etc. 1970s

Speed of processing in the human visual system

Simon Thorpe, Denis Fize & Catherine Marlot

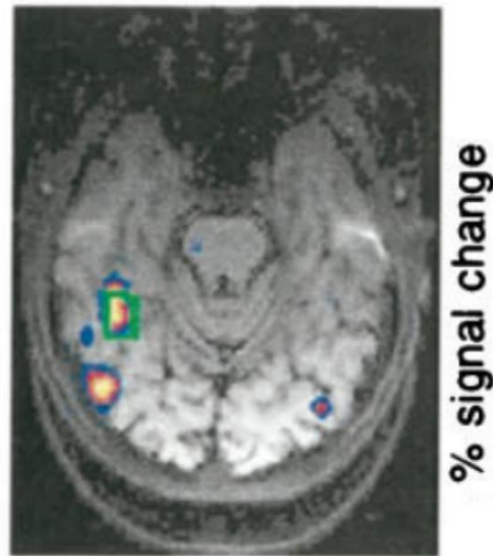


Thorpe, et al. *Nature*, 1996

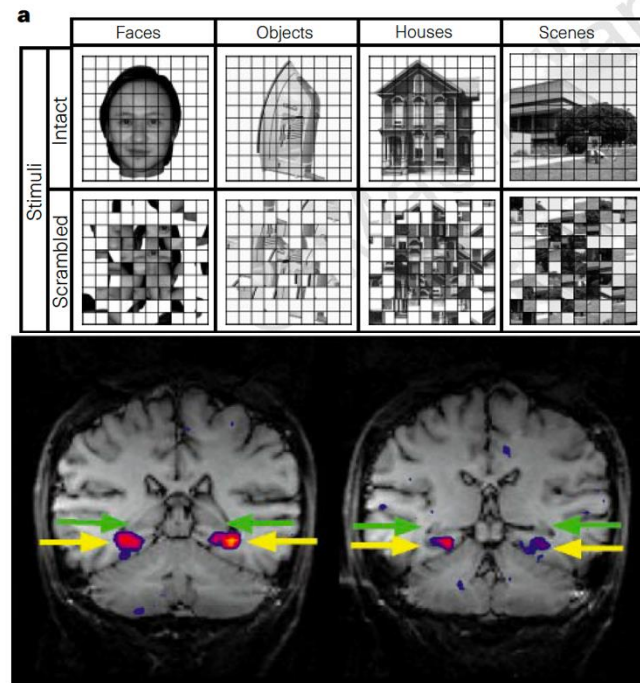
April 1, 2025

Neural correlates of object & scene recognition

Faces > Houses

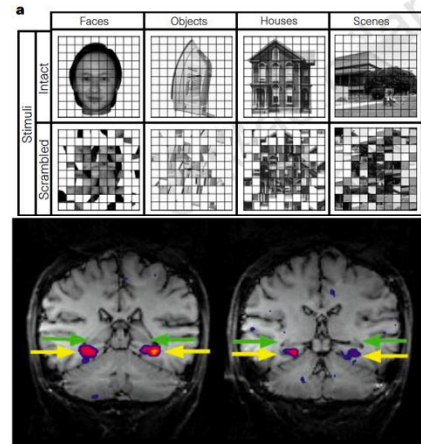


Kanwisher et al. J. Neuro. 1997

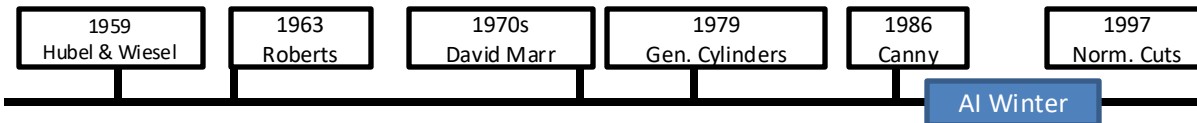


Epstein & Kanwisher, Nature, 1998

Visual recognition is a fundamental task for visual intelligence



Recognition via Grouping (1990s)



Normalized Cuts, Shi and Malik, 1997

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Slide inspiration: Justin Johnson

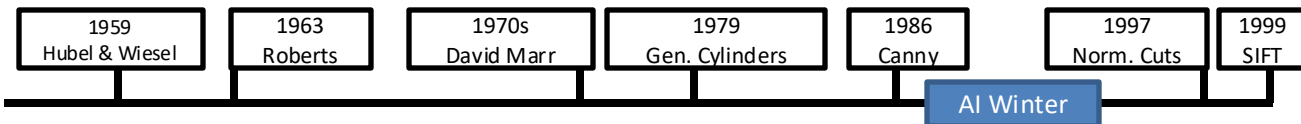
Recognition via Matching (2000s)



Image is public domain



Image is public domain



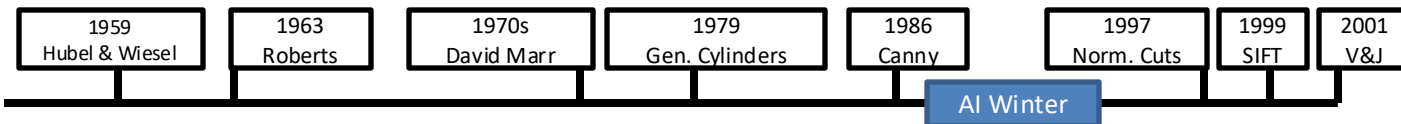
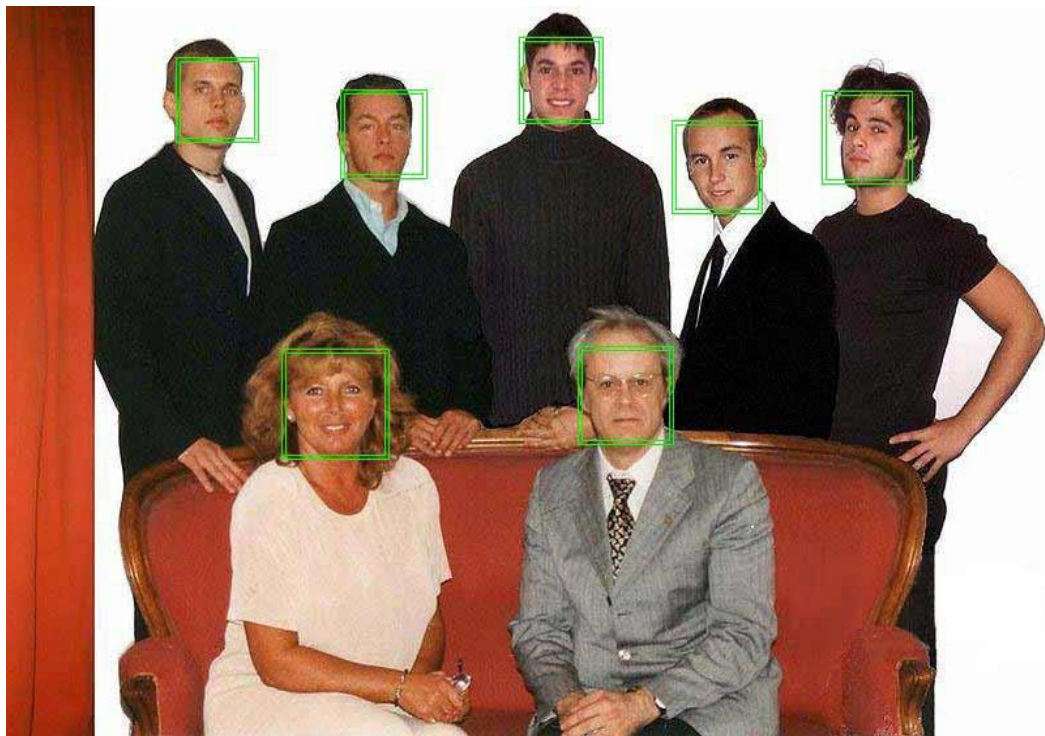
SIFT, David
Lowe, 1999

Slide inspiration: Justin Johnson

Face Detection

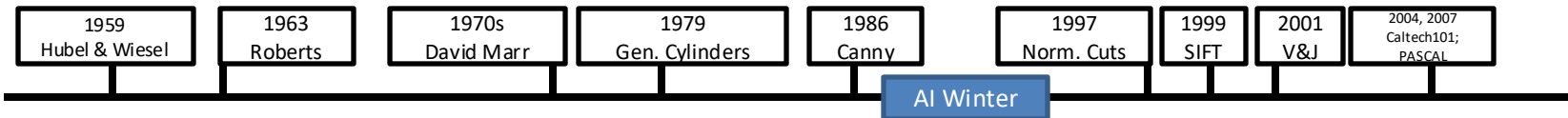
Viola and Jones, 2001

One of the first successful applications of machine learning to vision



Slide inspiration: Justin Johnson

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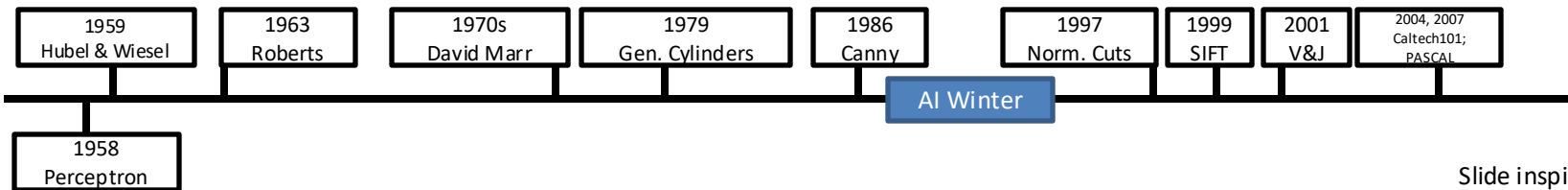
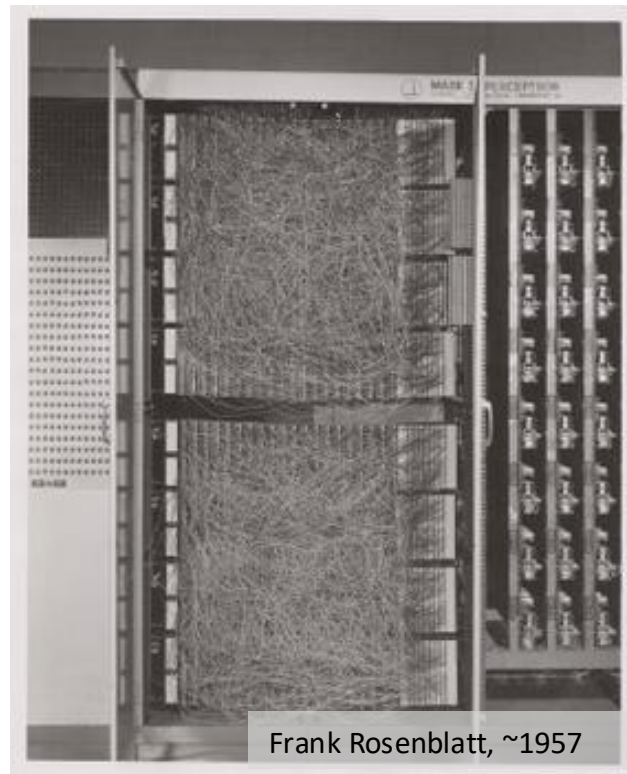
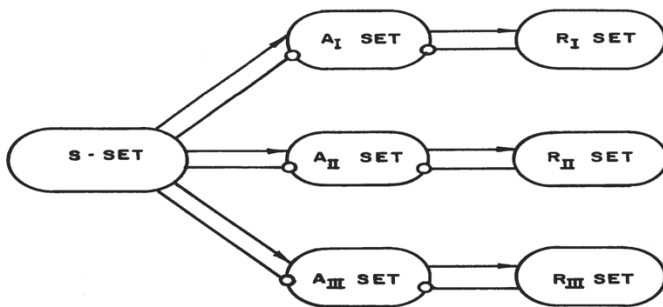
Fei-Fei Li & Ehsan Adeli & Zane Durante

Learning representations by back-propagating errors

David E. Rumelhart*, Geoffrey E. Hinton†
& Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California,
San Diego, La Jolla, California 92093, USA

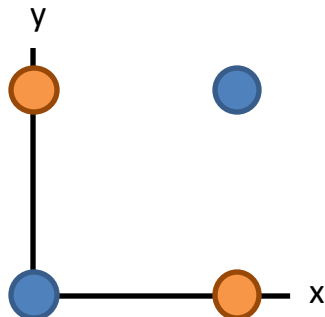
† Department of Computer Science, Carnegie-Mellon University,
Pittsburgh, Philadelphia 15213, USA



Slide inspiration: Justin Johnson

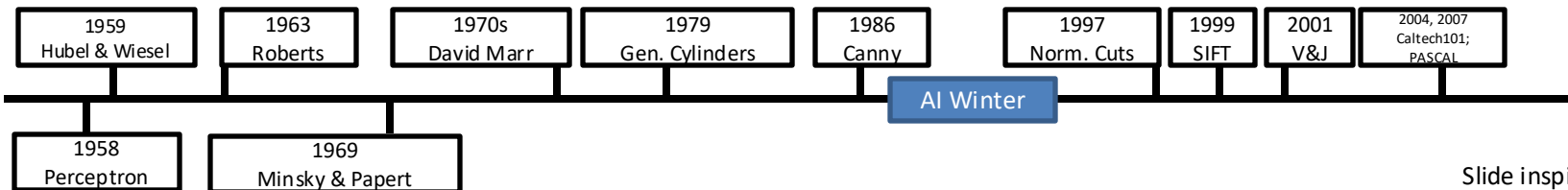
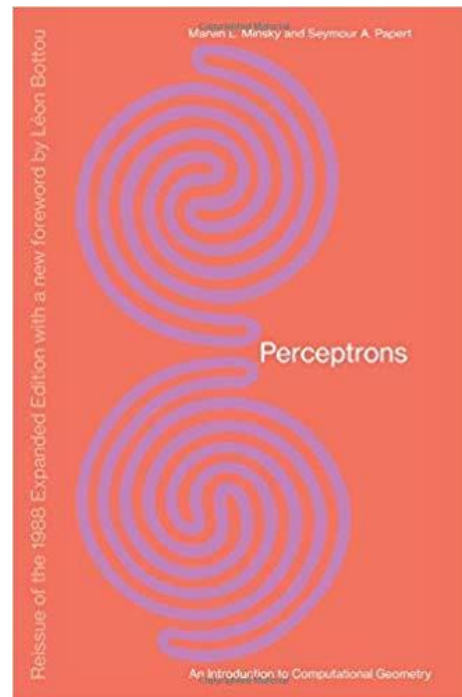
Minsky and Papert, 1969

X	Y	$F(x,y)$
0	0	0
0	1	1
1	0	1
1	1	0



Shown that Perceptrons could not learn the XOR function

Caused a lot of disillusionment in the field



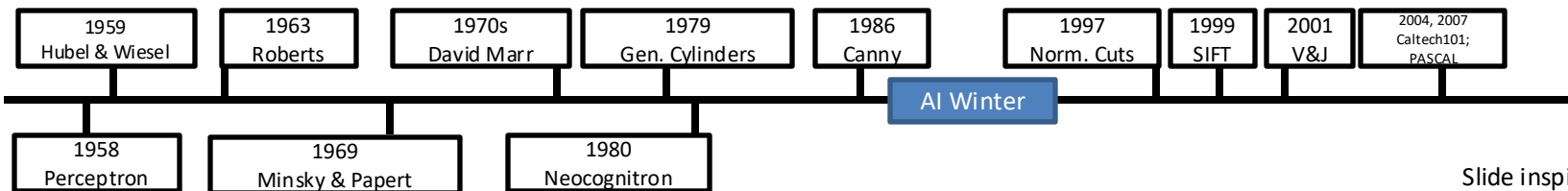
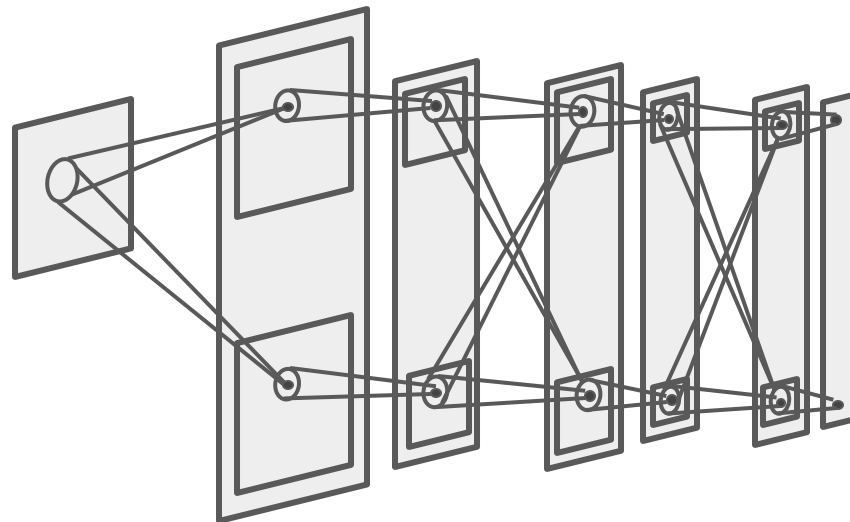
Slide inspiration: Justin Johnson

Neocognitron: Fukushima, 1980

Computational model the visual system,
directly inspired by Hubel and Wiesel's
hierarchy of complex and simple cells

Interleaved simple cells (convolution)
and complex cells (pooling)

No practical training algorithm

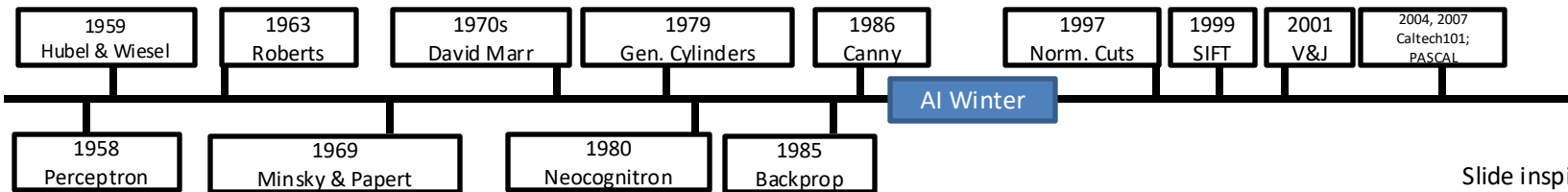
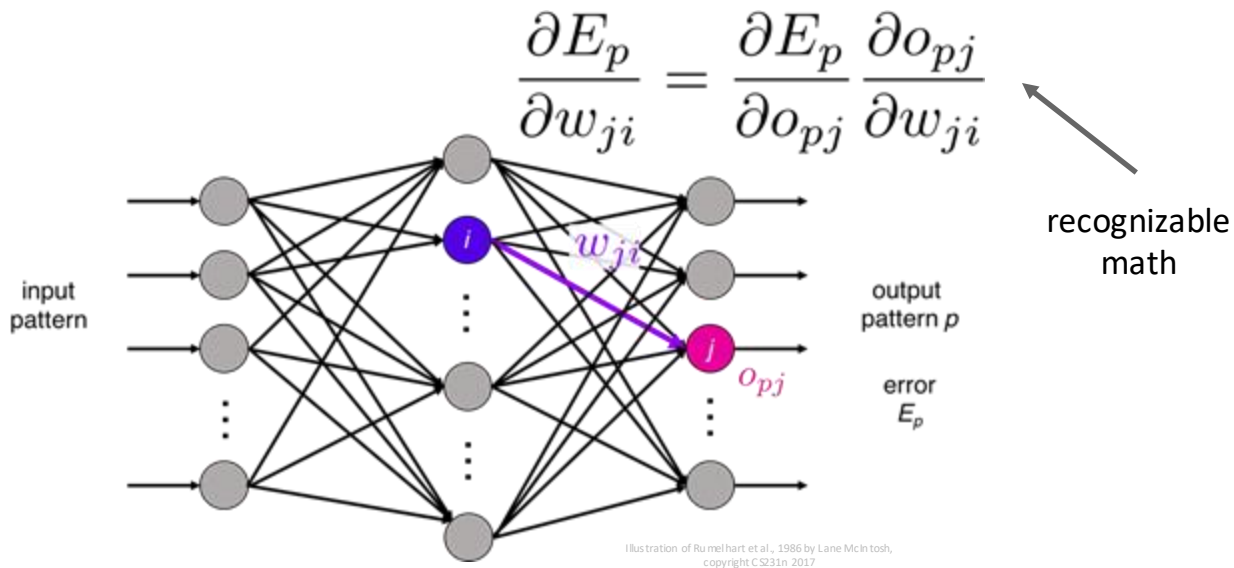


Slide inspiration: Justin Johnson

Backprop: Rumelhart, Hinton, and Williams, 1986

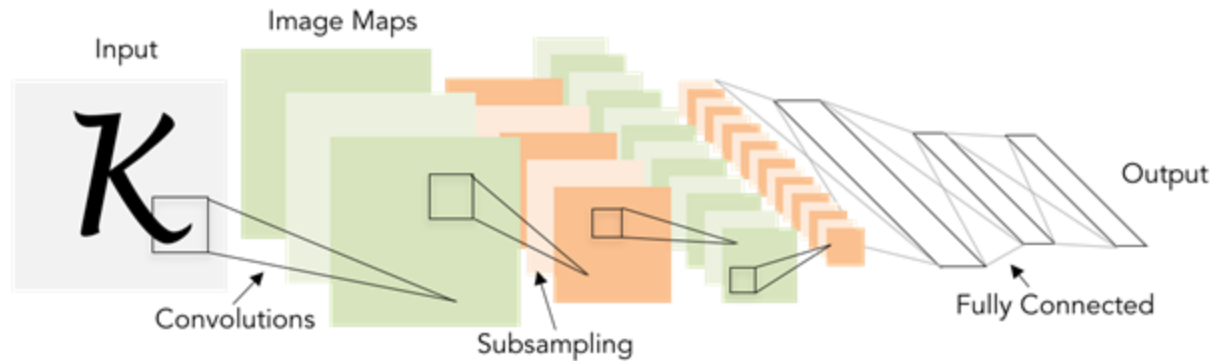
Introduced
backpropagation
for computing
gradients in neural
networks

Successfully trained
perceptrons with
multiple layers



Slide inspiration: Justin Johnson

Convolutional Networks: LeCun et al, 1998

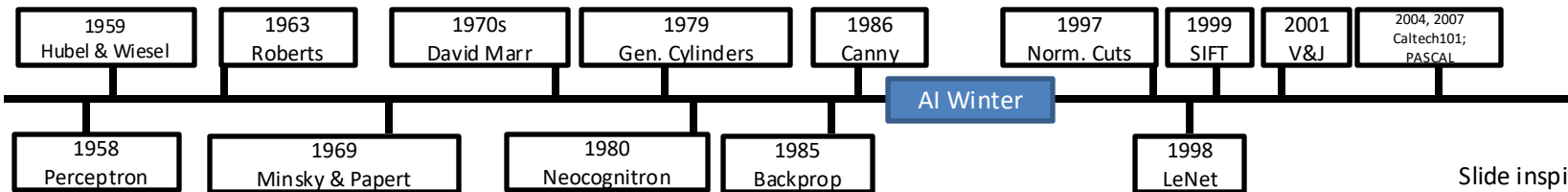


Applied backprop algorithm to a Neocognitron-like architecture

Learned to recognize handwritten digits

Was deployed in a commercial system by NEC, processed handwritten checks

Very similar to our modern convolutional networks!



Slide inspiration: Justin Johnson

2000s: “Deep Learning”

People tried to train neural networks that were deeper and deeper

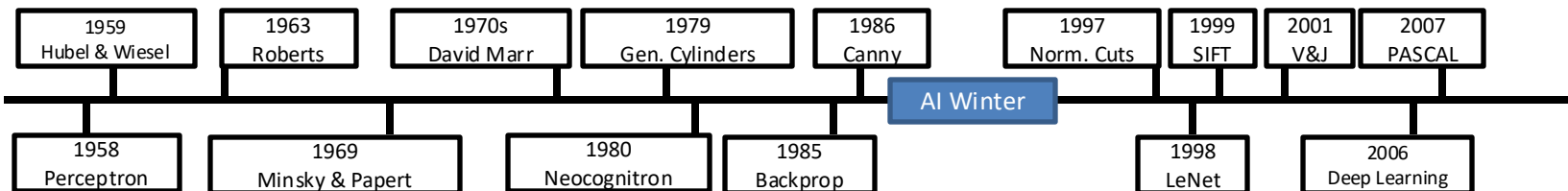
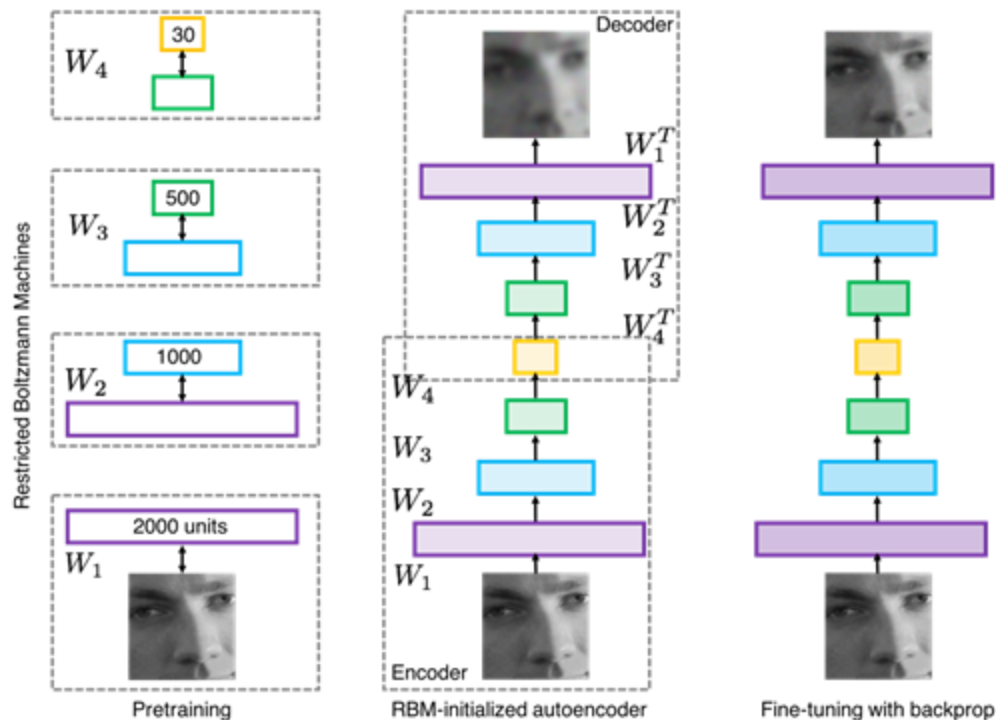
Not a mainstream research topic at this time

Hinton and Salakhutdinov, 2006

Bengio et al, 2007

Lee et al, 2009

Glorot and Bengio, 2010



2000s: “Deep Learning”

People tried to train neural networks that were deeper and deeper

Not a mainstream research topic at this time

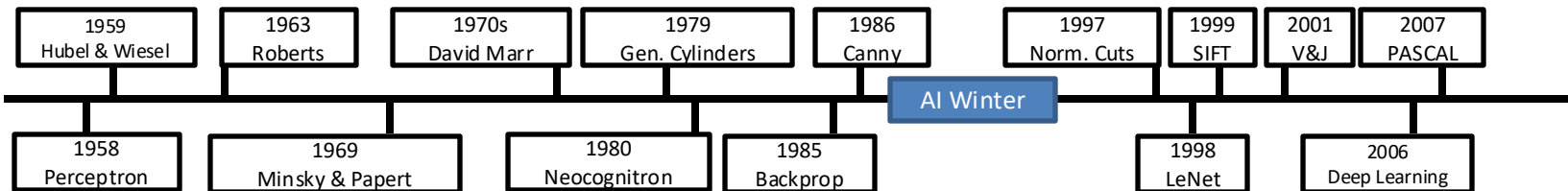
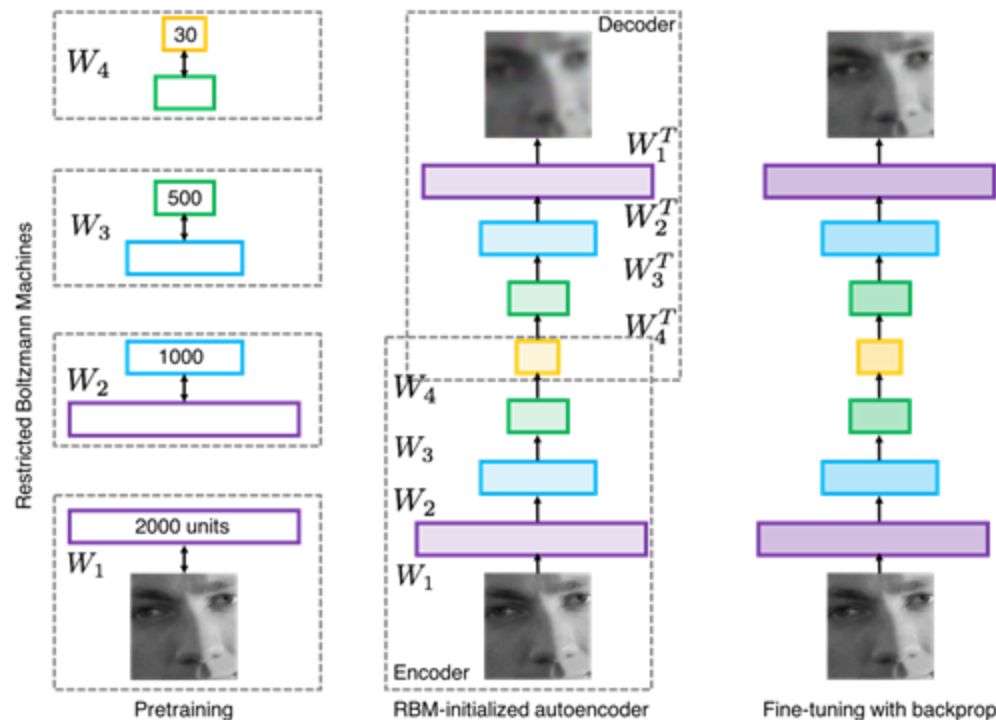
No good dataset to work on

Hinton and Salakhutdinov, 2006

Bengio et al, 2007

Lee et al, 2009

Glorot and Bengio, 2010



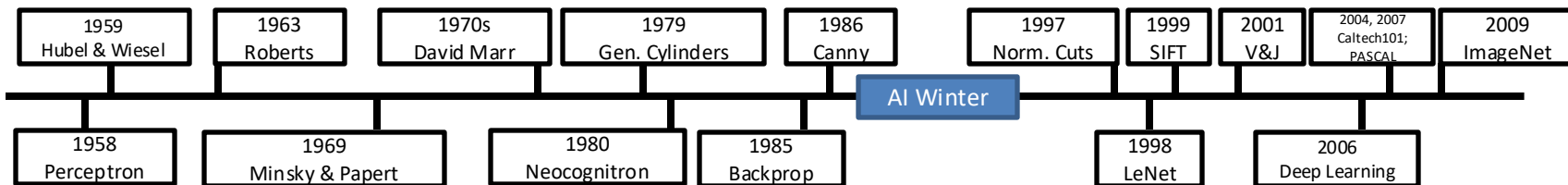
IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:
1,000 object classes
1,431,167 images

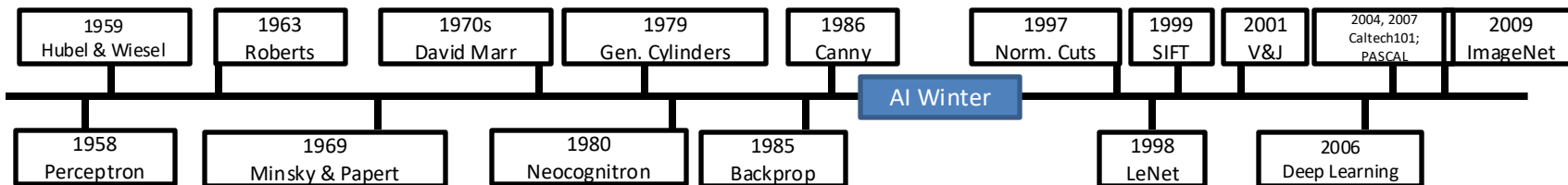
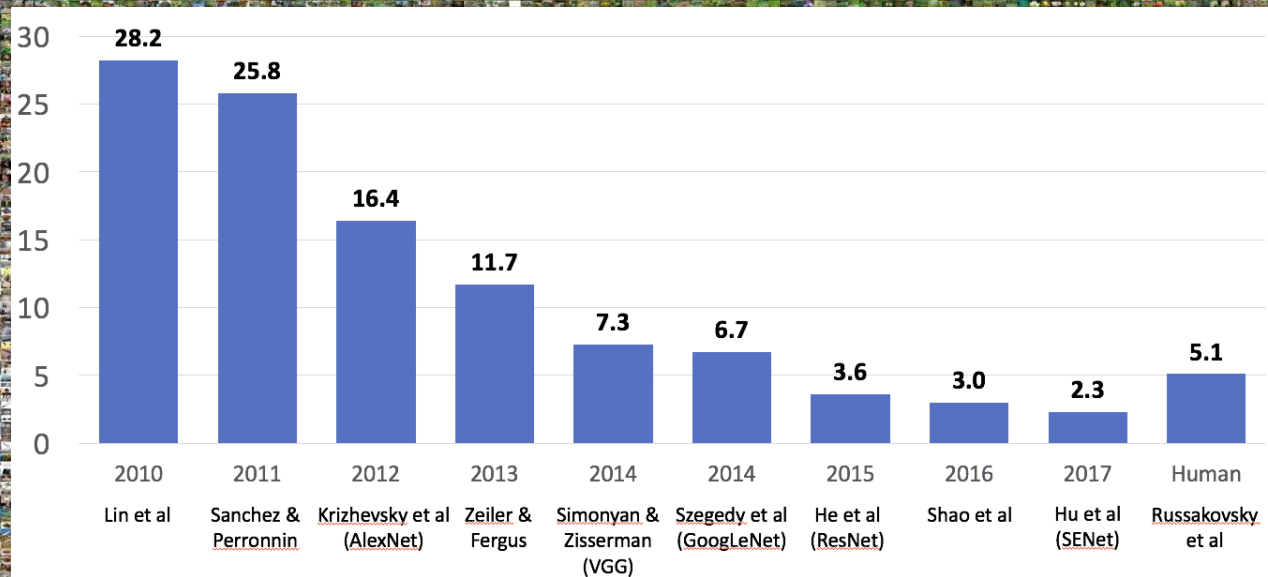


Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

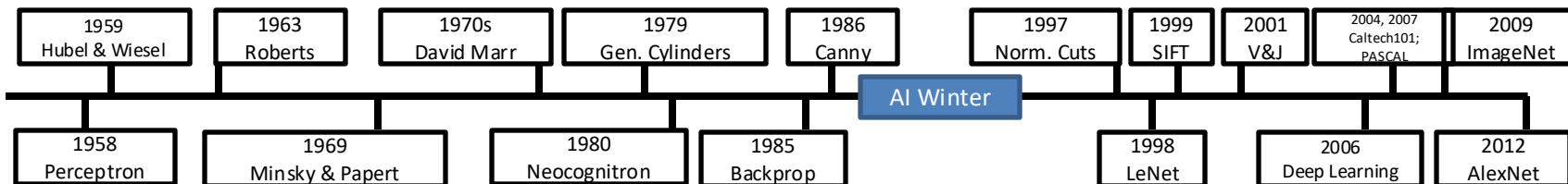
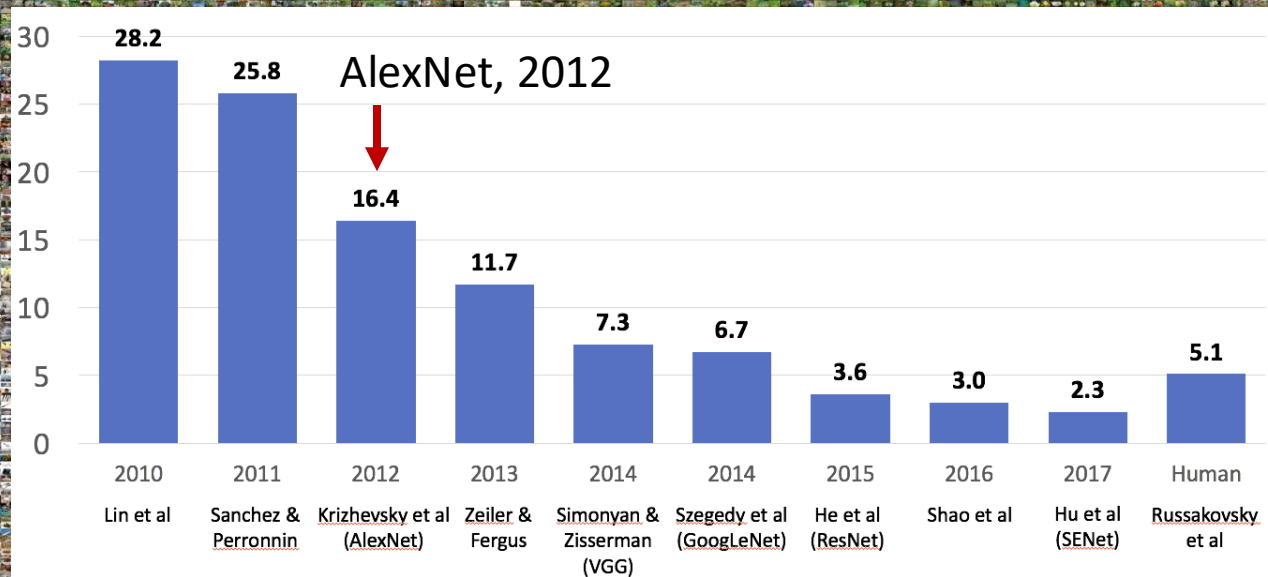
Deng et al, 2009
Russakovsky et al. IJCV 2015



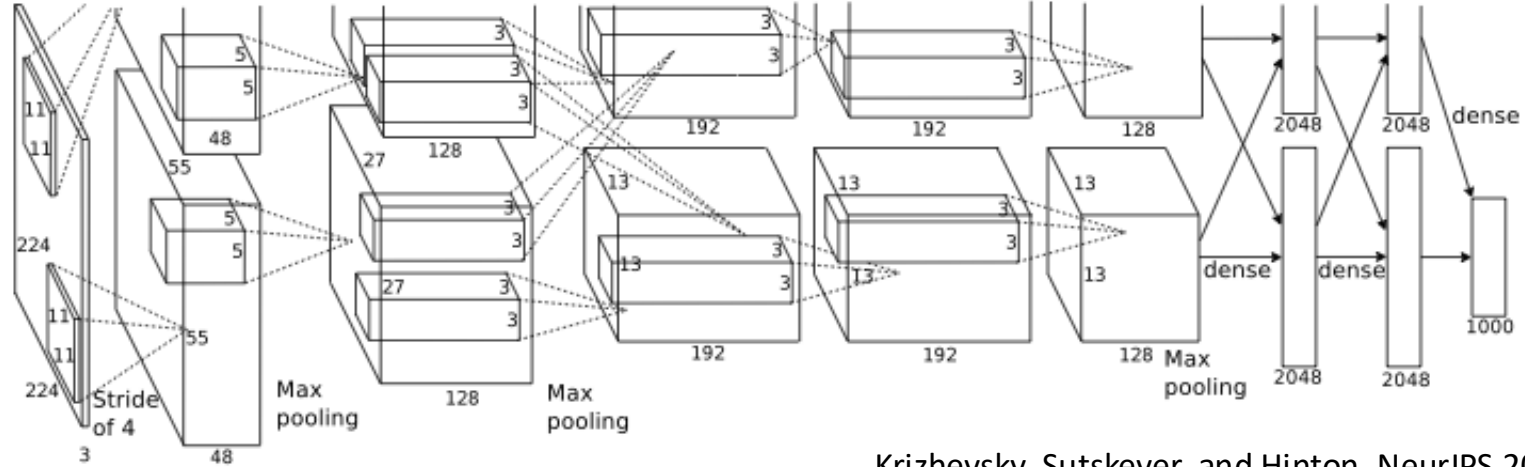
IMAGENET Large Scale Visual Recognition Challenge



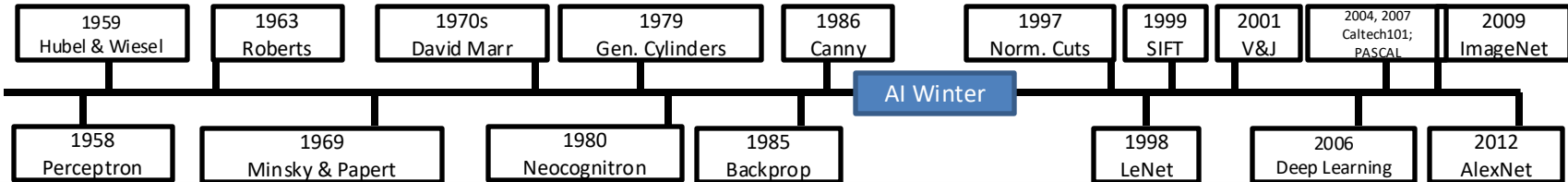
IMAGENET Large Scale Visual Recognition Challenge



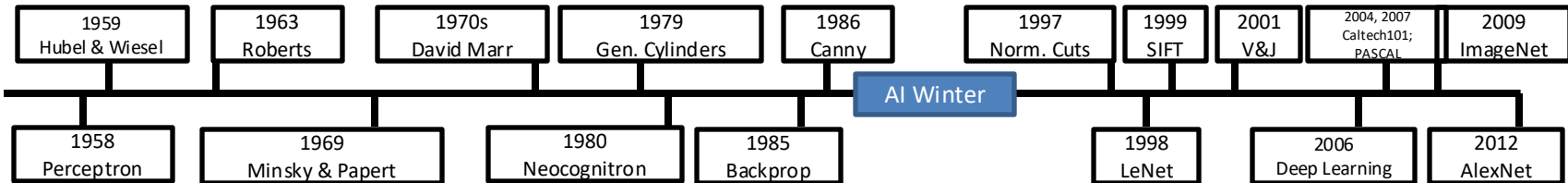
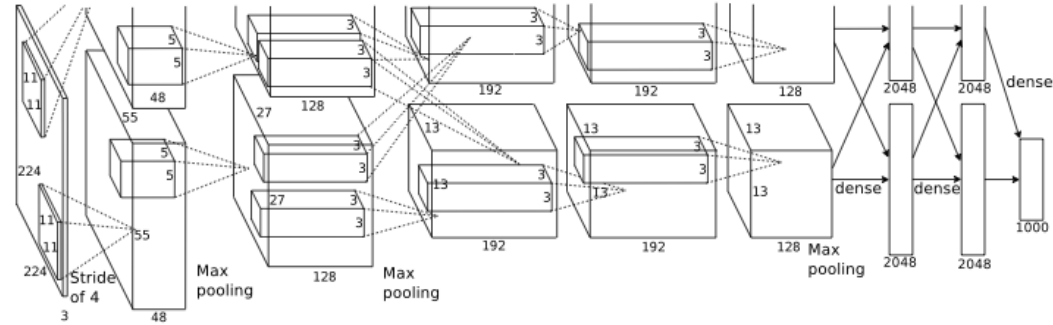
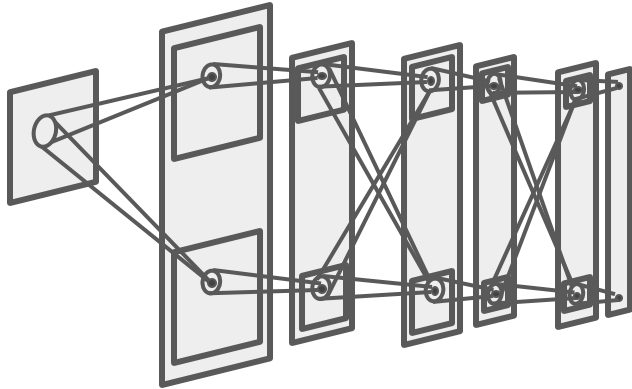
AlexNet: Deep Learning Goes Mainstream



Krizhevsky, Sutskever, and Hinton, NeurIPS 2012

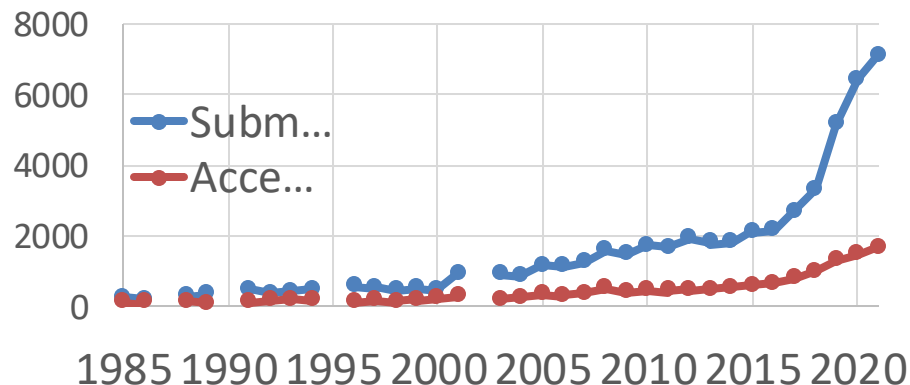


AlexNet vs. Neocognitron: 32 years apart

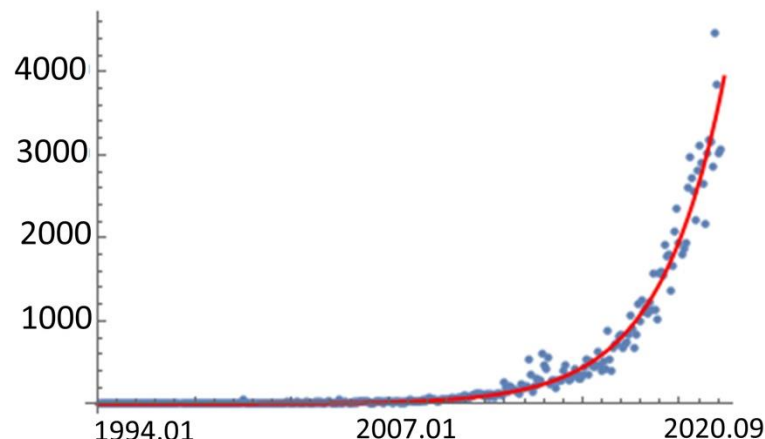


2012 to Present: Deep Learning Explosion

CVPR Papers

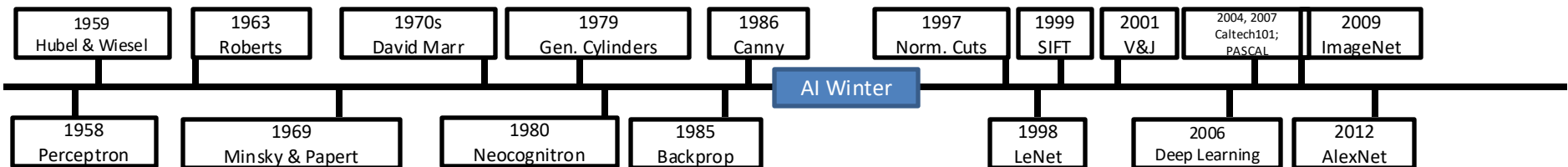


ML+AI arXiv papers per month



Publications at top Computer Vision conference

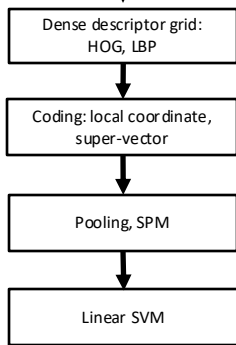
arXiv papers per month [\(source\)](#)



2012 to Present: Deep Learning is Everywhere

Year 2010

NEC-UIUC

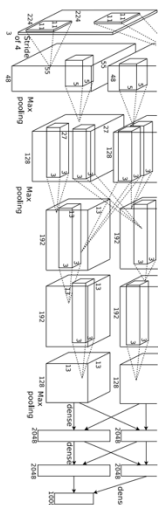


[Lin CVPR 2011]

Lion image by Swiss frog
is
licensed under CC BY 3.0

Year 2012

SuperVision

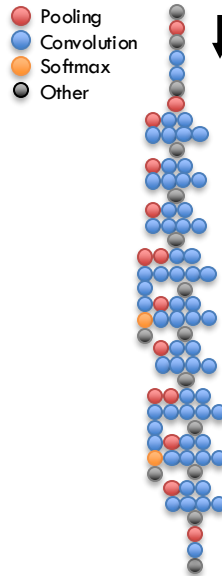


[Krizhevsky NIPS 2012]

Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

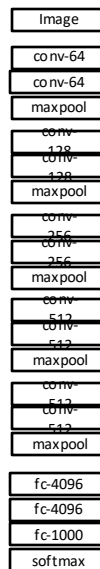
Year 2014

GoogLeNet



[Szegedy arxiv 2014]

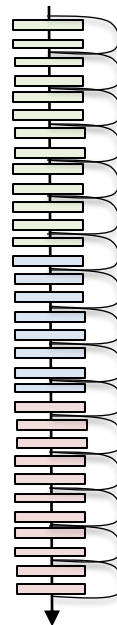
VGG



[Simonyan arxiv 2014]

Year 2015

MSRA



[He ICCV 2015]

2012 to Present: Deep Learning is Everywhere

Image Classification



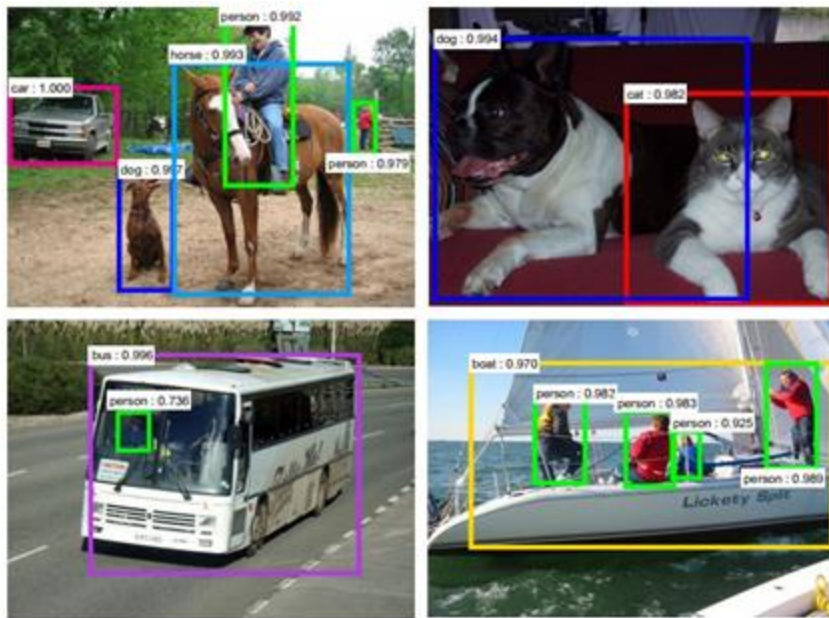
Image Retrieval



Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

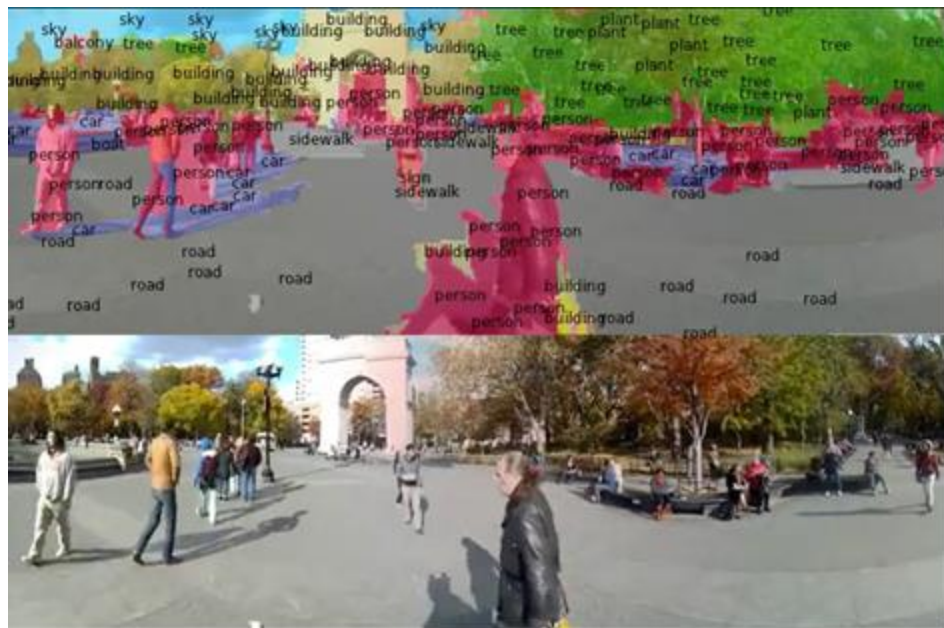
2012 to Present: Deep Learning is Everywhere

Object Detection



Ren, He, Girshick, and Sun, 2015

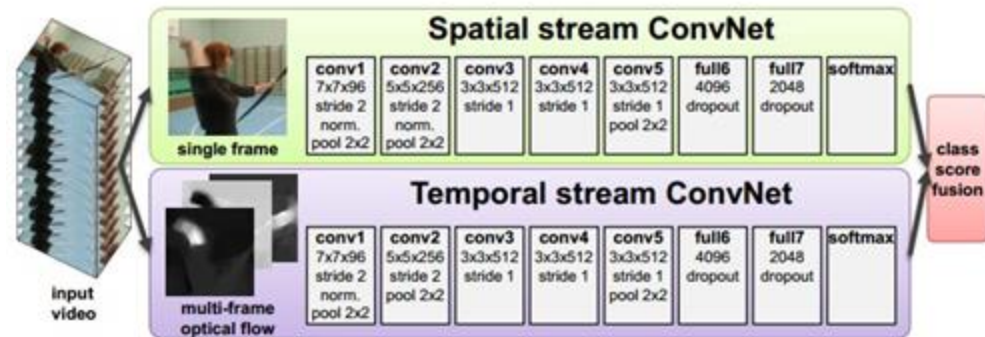
Image Segmentation



Fabaret et al, 2012

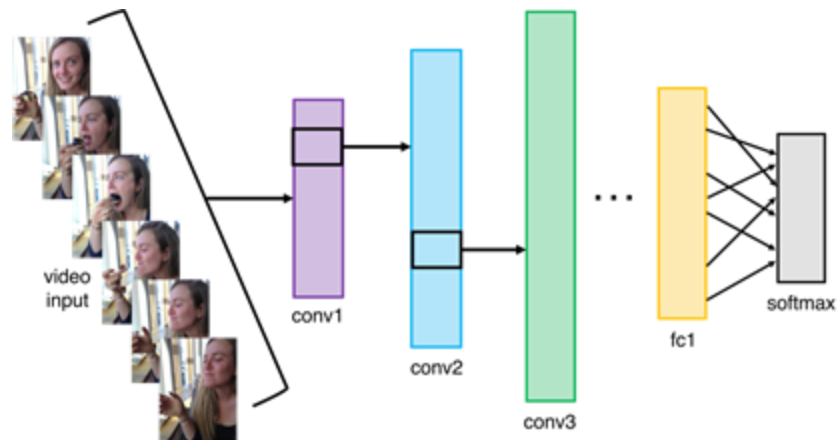
2012 to Present: Deep Learning is Everywhere

Video Classification



Simonyan et al, 2014

Activity Recognition

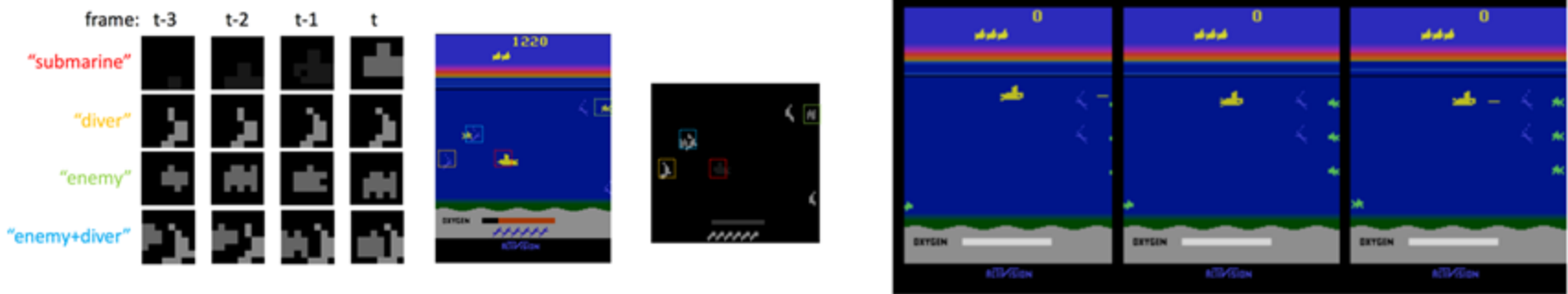


2012 to Present: Deep Learning is Everywhere

Pose Recognition (Toshev and Szegedy, 2014)

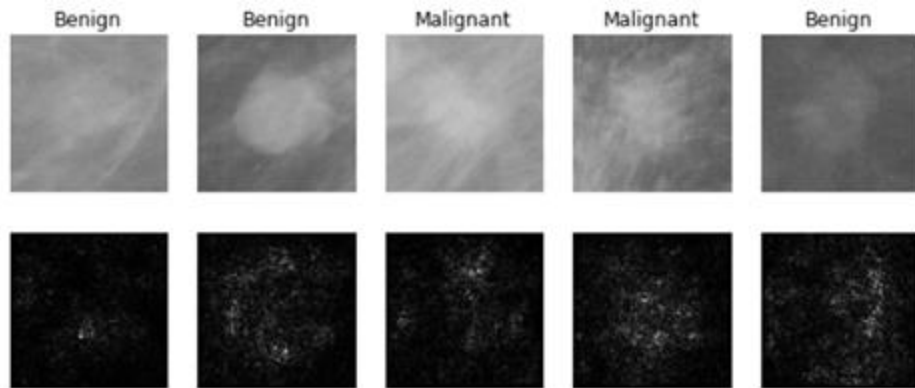


Playing Atari games (Guo et al, 2014)



2012 to Present: Deep Learning is Everywhere

Medical Imaging



Levy et al, 2016 Figure reproduced with permission

Galaxy Classification



Dieleman et al, 2014

From left to right: [public domain by NASA](#), usage permitted by ESA/Hubble, [public domain by NASA](#), and [public domain](#).

Whale recognition



Kaggle Challenge

[This image](#) by Christin Khan is in the public domain and originally came from the U.S. NOAA.

2012 to Present: Deep Learning is Everywhere



*A white teddy bear
sitting in the grass*



*A man in a baseball
uniform throwing a ball*



*A woman is holding
a cat in her hand*

Image Captioning

Vinyals et al, 2015

Karpathy and Fei-Fei,
2015



*A man riding a wave
on top of a surfboard*



*A cat sitting on a
suitcase on the floor*

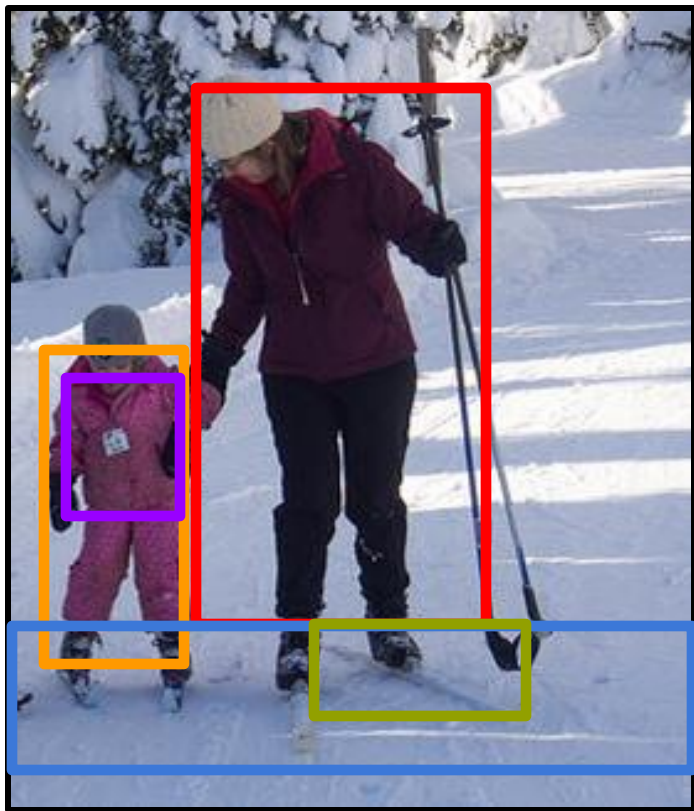


*A woman standing on a
beach holding a surfboard*

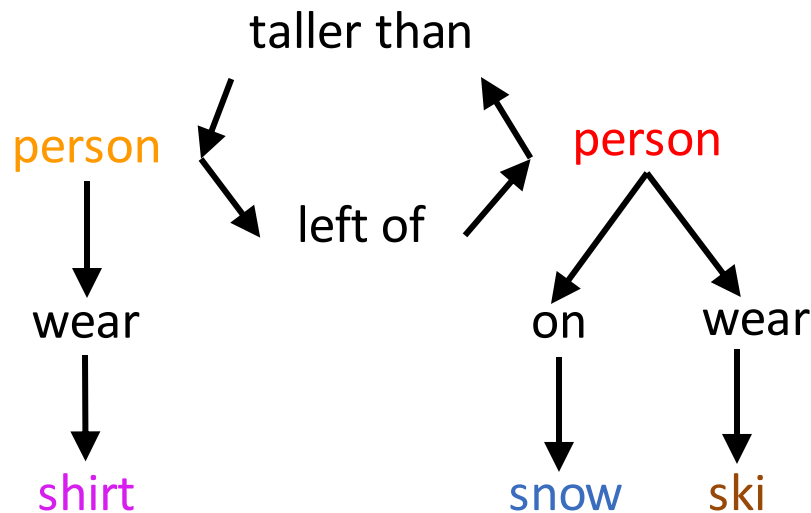
All images are CC0 Public domain:
<https://pixabay.com/en/luggage-antique-cat-1643010/>
<https://pixabay.com/en/teddy-plush-bears-cute-teddy-bear-1623436/>
<https://pixabay.com/en/surf-wave-surfer-sport-litoral-1668716/>
<https://pixabay.com/en/woman-female-model-sport-trail-adult-983967/>
<https://pixabay.com/en/handstand-lake-meditation-496008/>
<https://pixabay.com/en/baseball-player-shortstop-in-field-1045263/>

Captions generated by Justin Johnson using *NeuralTalk2*

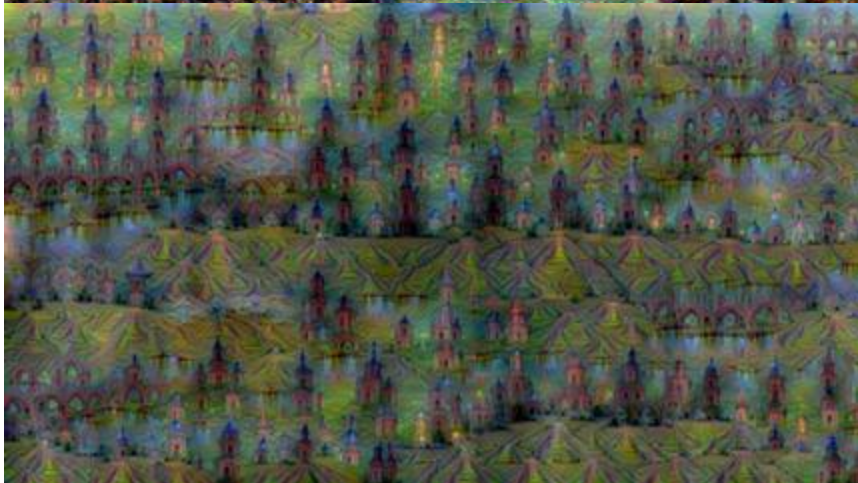
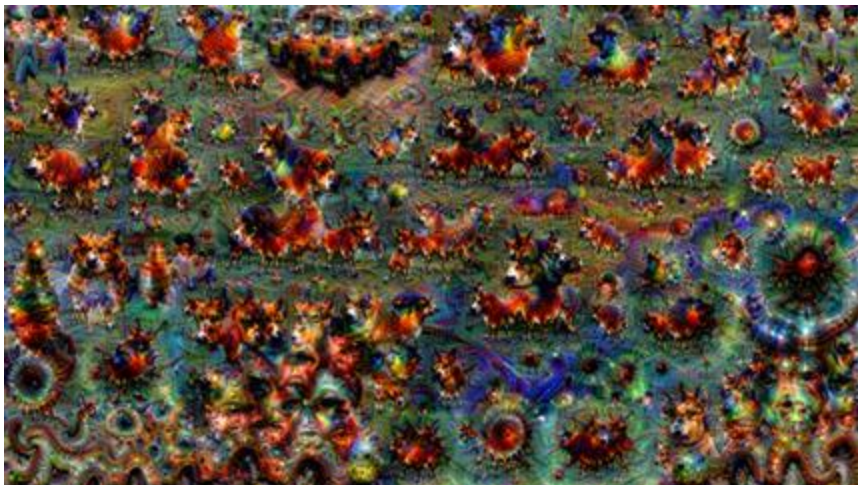
2012 to Present: Deep Learning is Everywhere



Results:
spatial, comparative, asymmetrical, verb,
prepositional



Krishna*, Lu*, Bernstein, Fei-Fei, *ECCV* 2016



Original image is CC0 public domain
 Starry Night and Tree Roots by Van Gogh are in the public domain
 Bokeh image is in the public domain
 Stylized images copyright Justin Johnson, 2015;
 reproduced with permission

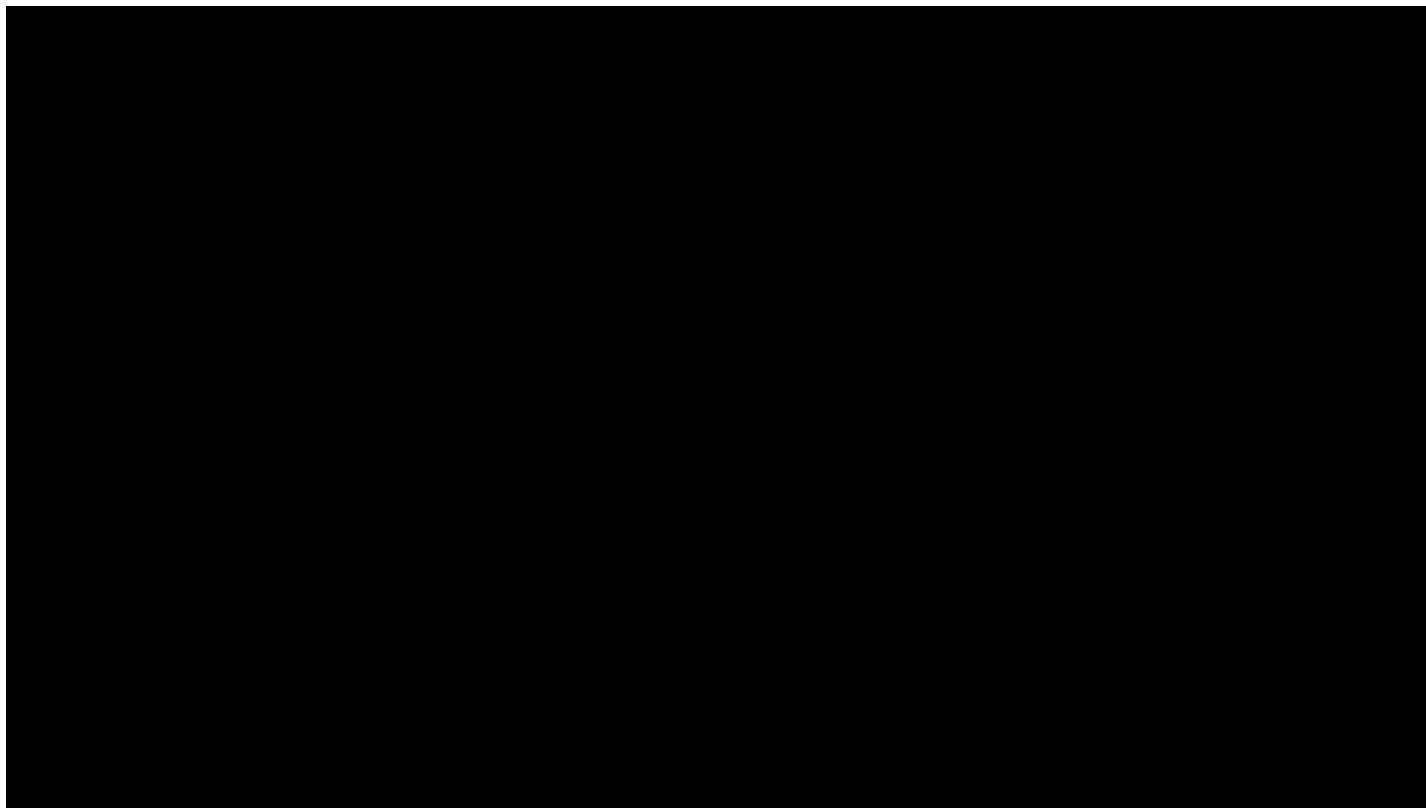


Mordvinsev et al, 2015
 Gatys et al, 2016

Figures copyright Justin Johnson, 2015. Reproduced with permission. Generated using the Inception v3 model from a blog post by Google Research.

Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere



Karras et al, "Progressive Growing of GANs for Improved Quality, Stability, and Variation", ICLR 2018

Slide inspiration: Justin Johnson

2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of an avocado. an armchair imitating an avocado.

AI-GENERATED IMAGES



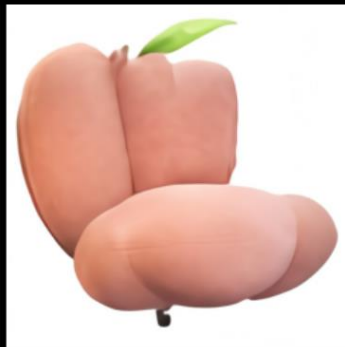
Slide inspiration: Justin Johnson

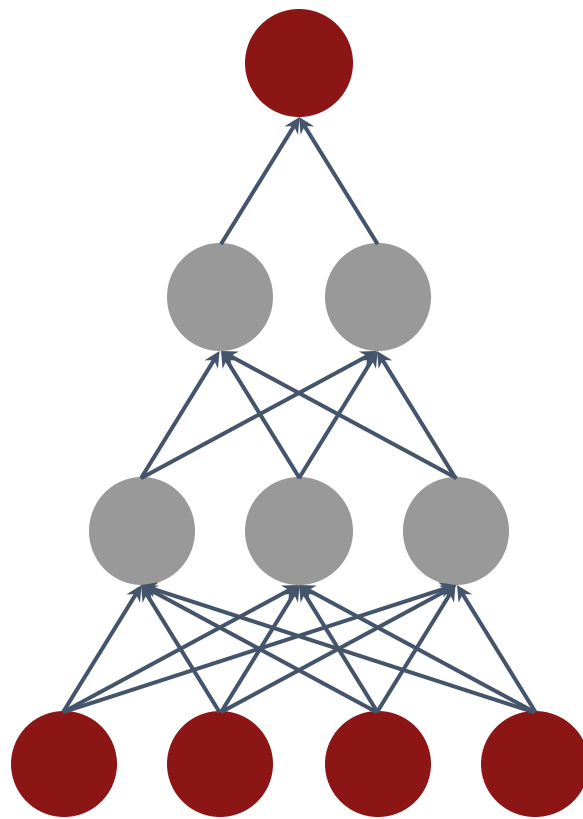
2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of a peach. an armchair imitating a peach.

AI-GENERATED IMAGES



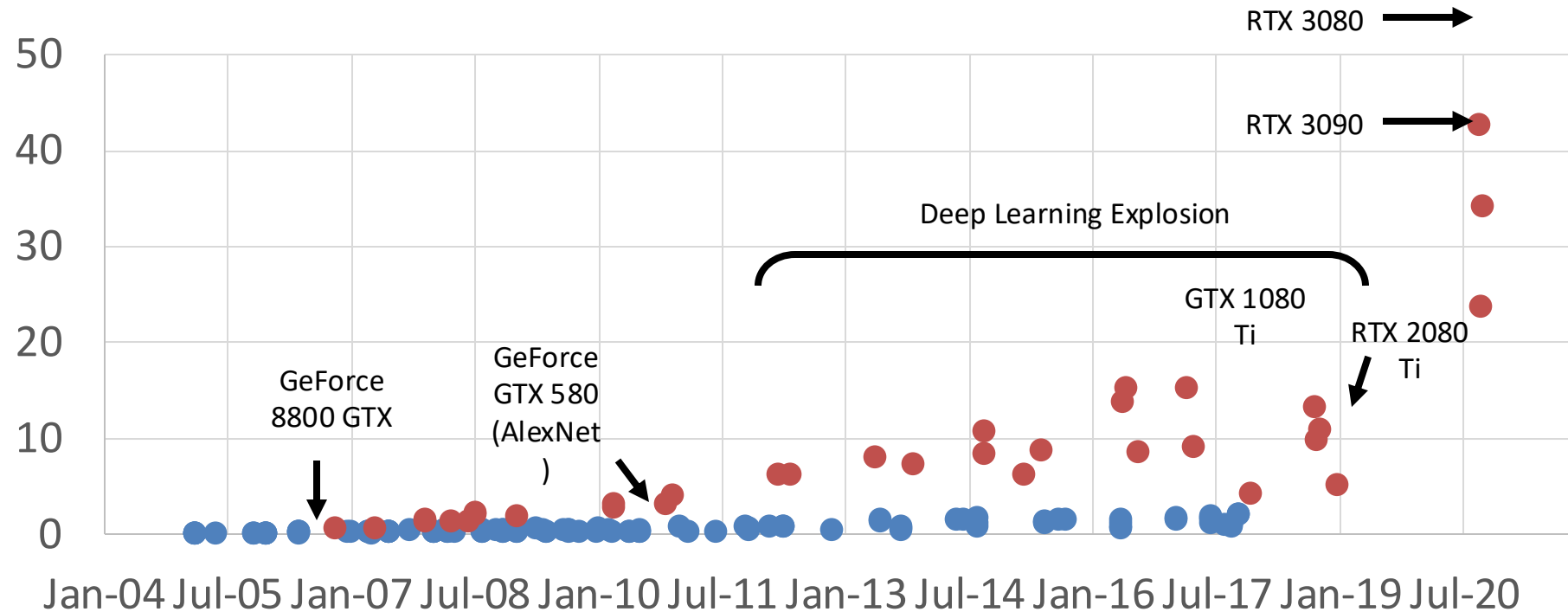


Algorithms



GFLOP per Dollar

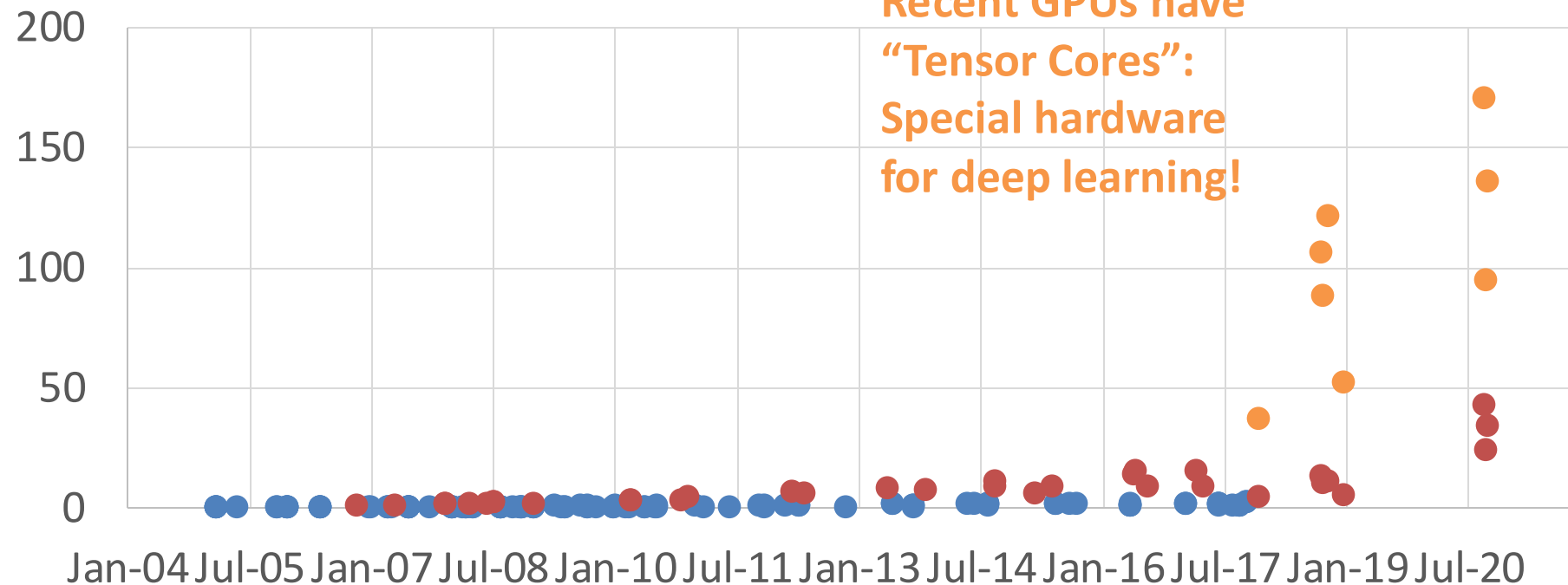
● CPU ● GPU (FP32)



GFLOP per Dollar

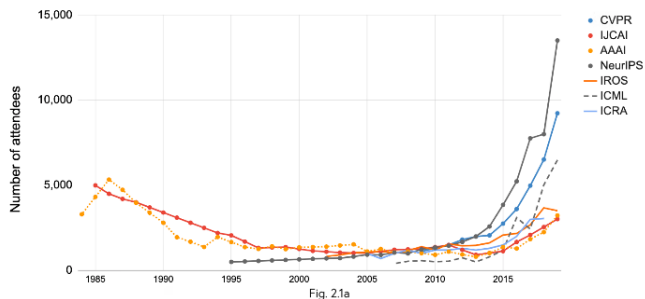
● CPU ● GPU (FP32) ● GPU (Tensor Core)

Recent GPUs have
“Tensor Cores”:
Special hardware
for deep learning!



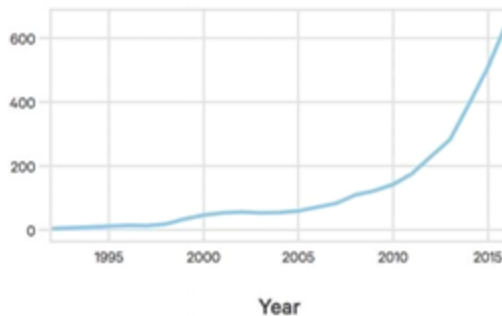
AI's Explosive Growth & Impact

Attendance at large conferences (1984-2019)
Source: Conference provided data.



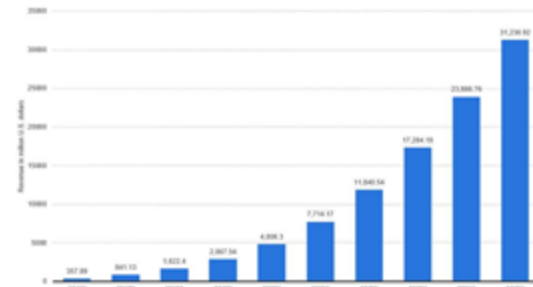
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

Source: Crunchbase, VentureSource, Sand
Hill Econometrics



**Enterprise Application AI
Revenue**

Source: Statista

Despite the successes, computer vision still has a long way to go

Computer Vision Can Cause Harm

Harmful Stereotypes

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8%
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8%
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4%

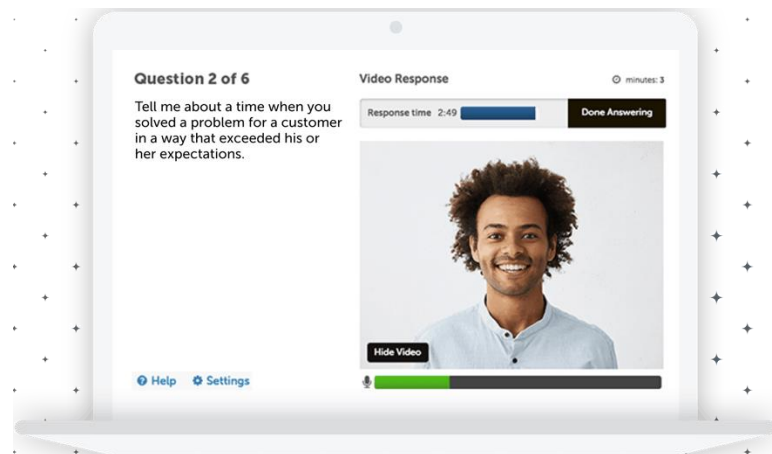


Affect people's lives

Technology

A face-scanning algorithm increasingly decides whether you deserve the job

HireVue claims it uses artificial intelligence to decide who's best for a job. Outside experts call it 'profoundly disturbing.'



Source: <https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decides-whether-you-deserve-job/>
<https://www.hirevue.com/platform/online-video-interviewing-software>

Example Credit: Timnit Gebru

Computer Vision Can Save Lives

How to take care of seniors while keeping them safe?



Early Symptom Detection of COVID-19



Monitor Patients with Mild Symptoms



Manage Chronic Conditions

Versatile



Mobility



Infection



Sleep



Diet



Scalable



Low-cost



Burden-free



And there is a lot we don't know how to do



https://fedandfit.com/wp-content/uploads/2020/06/summer-activities-for-kids_optimized-scaled.jpeg

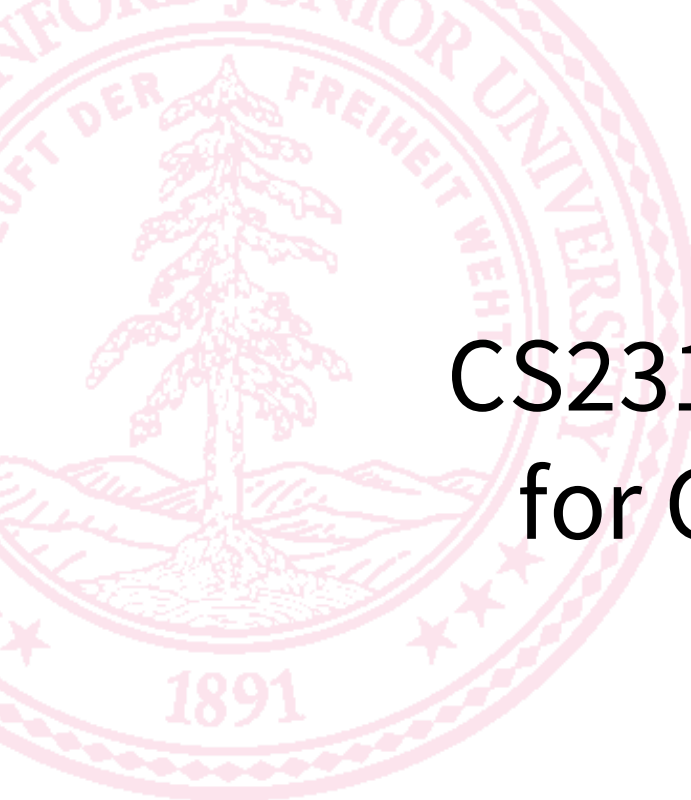


This image is
copyright-free [United States government work](#)

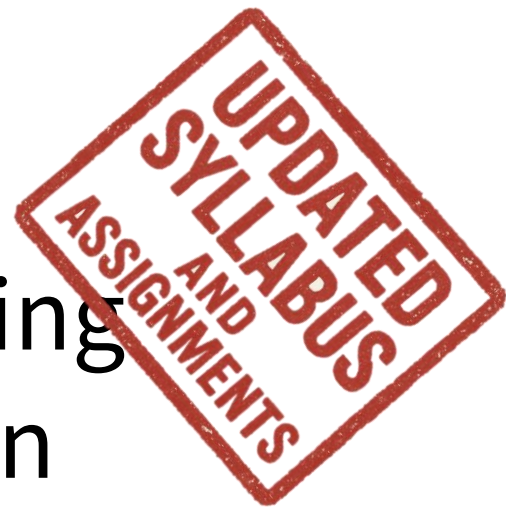
Slide inspiration: Andrej Karpathy

Today's agenda

- A brief history of computer vision & deep learning
- CS231n overview

The Stanford University seal is visible in the background on the left side of the slide. It features a red circular border with the text 'STANFORD UNIVERSITY' at the top and 'FUND. 1891' at the bottom. Inside the circle is a red tree (El Palo Alto) on a hill, with the words 'DIE LUFT DER FREIHEIT WEHT' (The wind of freedom blows) around it.

CS231n: Deep Learning for Computer Vision



Spring 2025

Lecture 1 – Part 2 – Overview

Instructors



Fei-Fei Li



Ehsan Adeli



Zane Durante



Justin Johnson

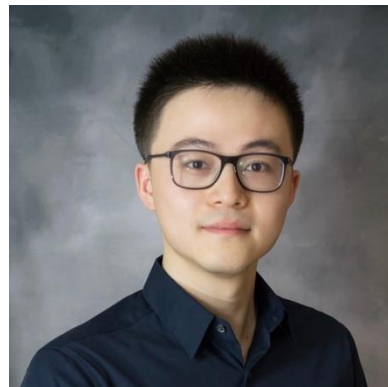
Guest Lecturers



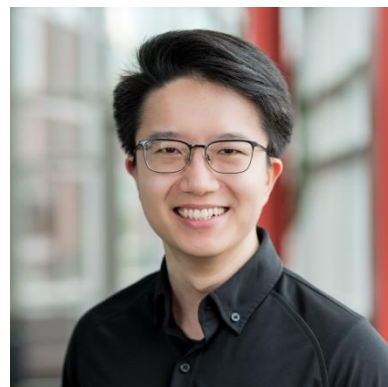
Jiajun Wu



Ranjay Krishna



Ruohan Gao



Yunzhu Li

CS231n overview

- Deep Learning Basics
- Perceiving and Understanding the Visual World
- Generative and Interactive Visual Intelligence
- Human-Centered Applications and Implications

Deep Learning Basics

- Image Classification: A core task in Computer Vision



This image by Nikita is
licensed under [CC-BY2.0](#)

→ cat

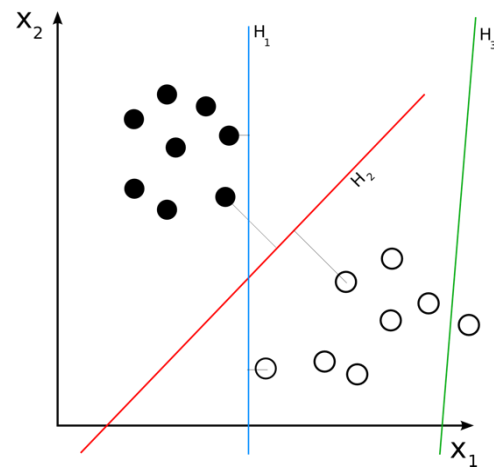
Deep Learning Basics

- Image Classification: A core task in Computer Vision



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→ cat



Linear Classifier

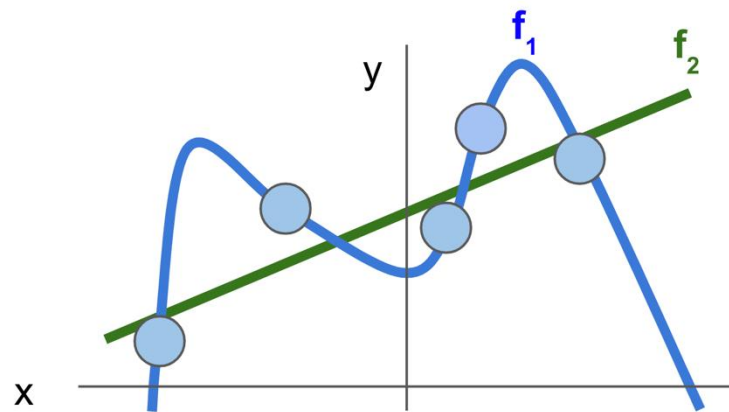
Deep Learning Basics

- Image Classification: A core task in Computer Vision



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→ cat



Regularization & Optimization

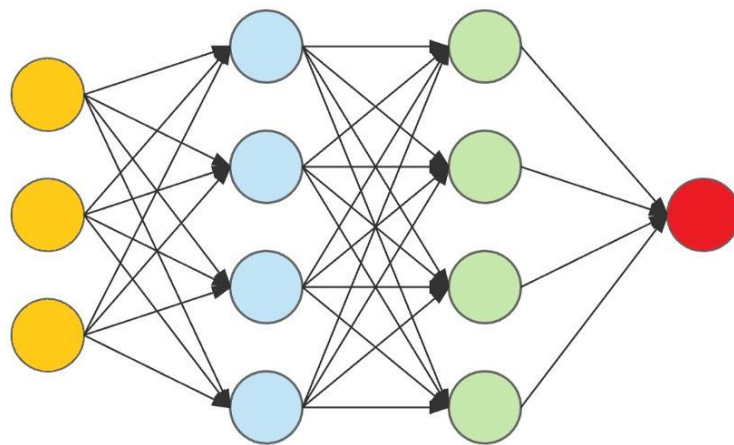
Deep Learning Basics

- Image Classification: A core task in Computer Vision



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Neural Networks

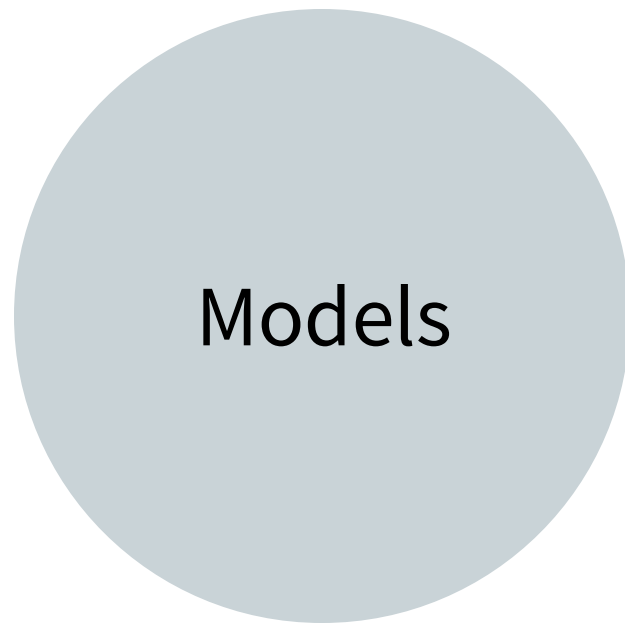
CS231n overview

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CS231n overview

- Deep Learning Basics
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Perceiving and Understanding the Visual World



Tasks Beyond Image Classification

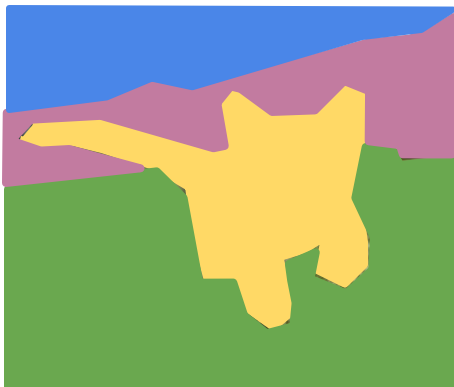
Classification



CAT

No spatial extent

Semantic Segmentation



GRASS, CAT, TREE,
SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



DOG, DOG, CAT

[This image is CC0 public domain](#)

Tasks Beyond Image Classification

Video Classification



Running? Jumping?

Multimodal Video Understanding



Visualization & Understanding



Models Beyond Multi-Layer Perceptron

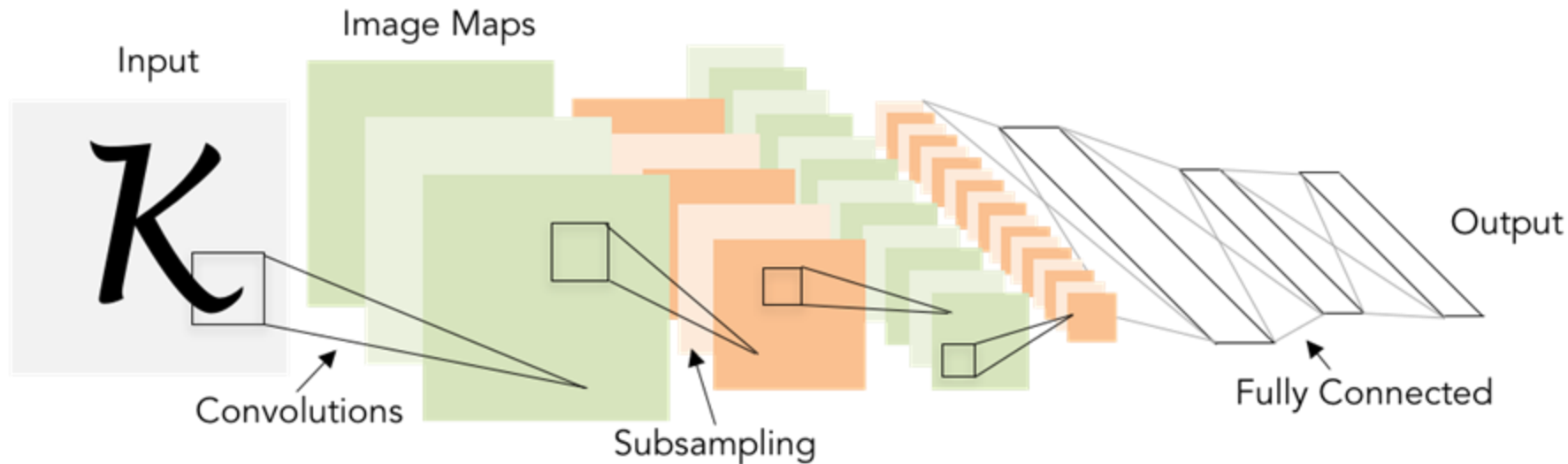
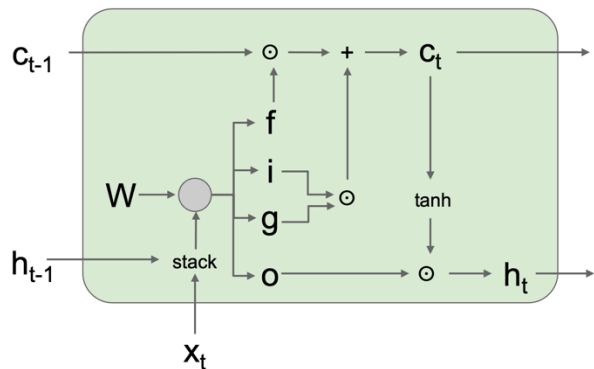
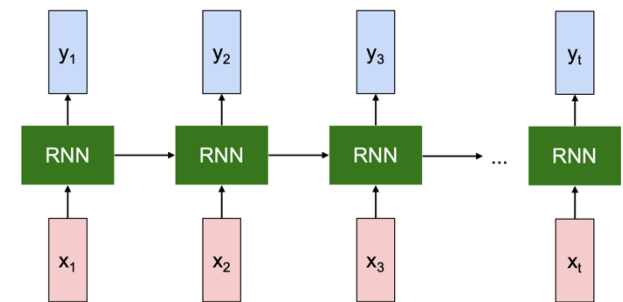


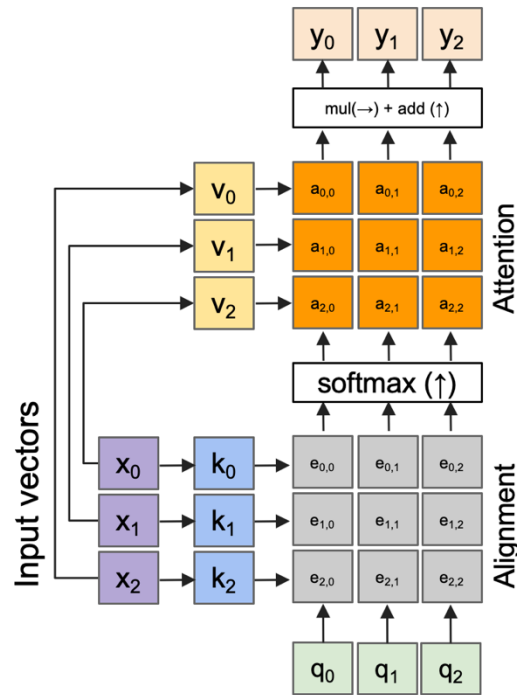
Illustration of LeCun et al. 1998 from CS231n 2017 Lecture 1

Convolutional neural network

Models Beyond Multi-Layer Perceptron



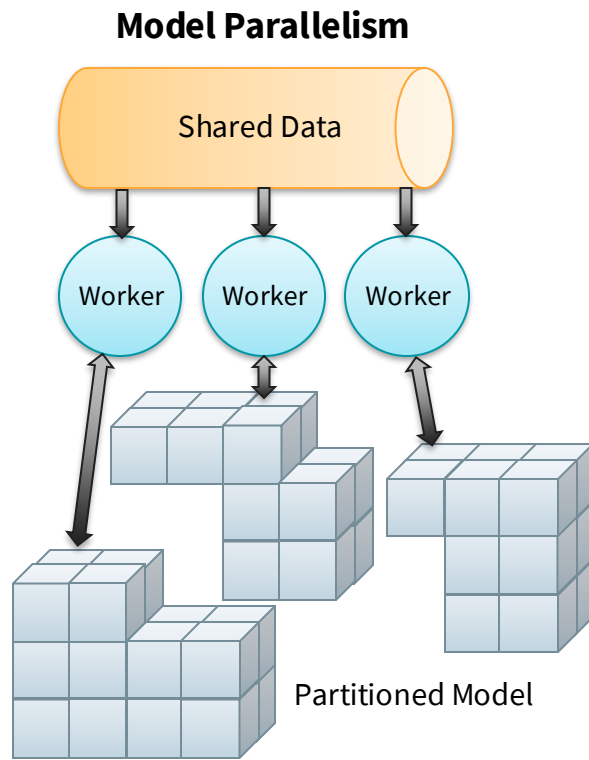
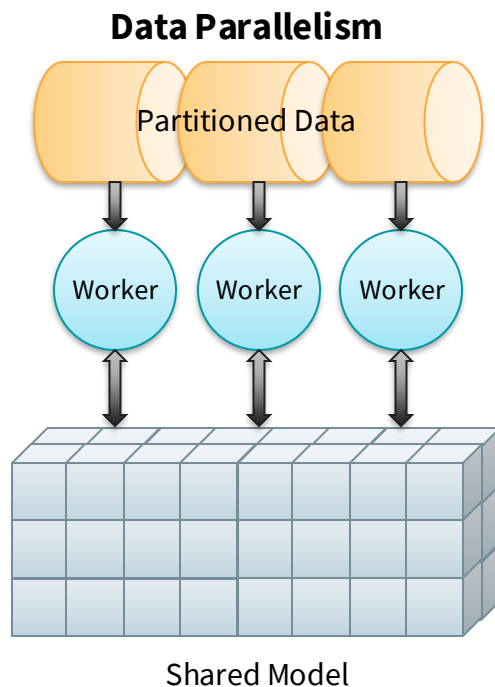
Recurrent neural network



Attention mechanism / Transformers

Large Scale Distributed Training

- Train Large Models on big datasets faster
- Scale beyond single GPU/machine limitations
- How?
 - Data Parallelism: Copy the model to all workers, split the data
 - Model Parallelism: Split model across devices
 - Synchronous vs. Asynchronous gradient updates



CS231n overview

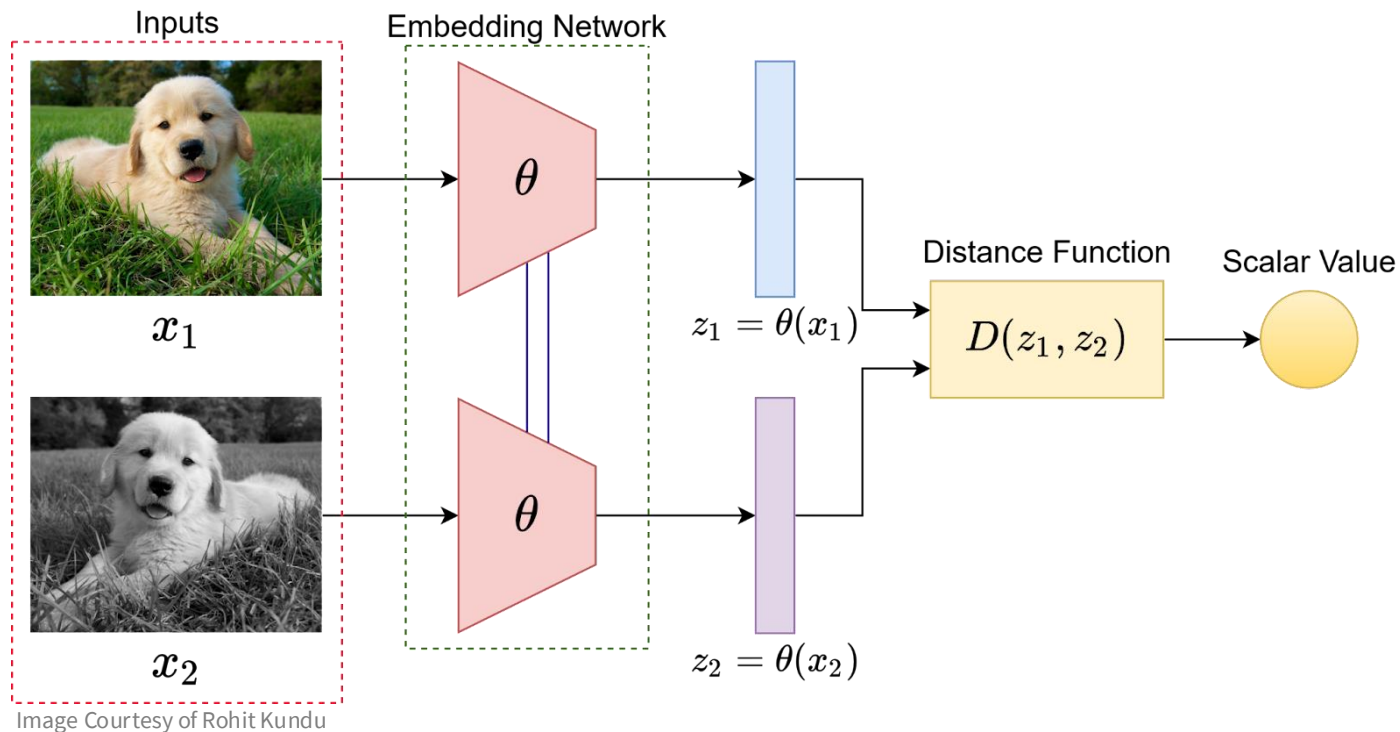
- Deep Learning Basics
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- Human-Centered Applications and Implications

CS231n overview

- Deep Learning Basics
- Perceiving and Understanding the Visual World
- **Generative and Interactive Visual Intelligence**
- Human-Centered Applications and Implications

Beyond 2D Recognition

Beyond 2D Recognition: Self-supervised Learning



Beyond 2D Recognition: Generative Modeling



Style Transfer

Beyond 2D Recognition: Generative Modeling



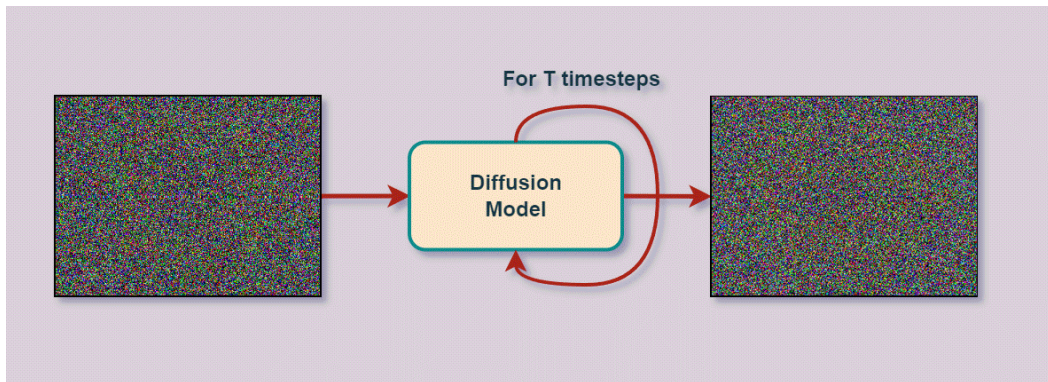
[This image is public domain](#)

“Teddy bears working on new AI research underwater with 1990s technology”

DALL-E 2

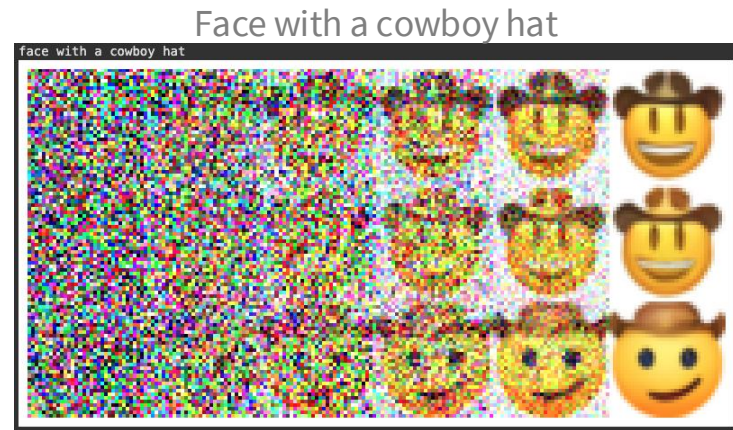
Beyond 2D Recognition: Generative Modeling

Image Generation using Diffusion Models

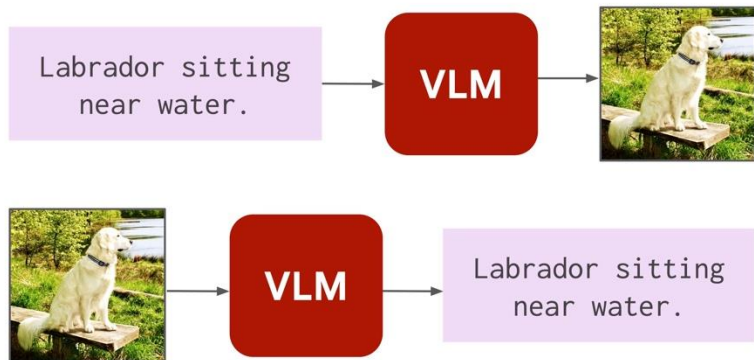


<https://learnopencv.com/image-generation-using-diffusion-models/>

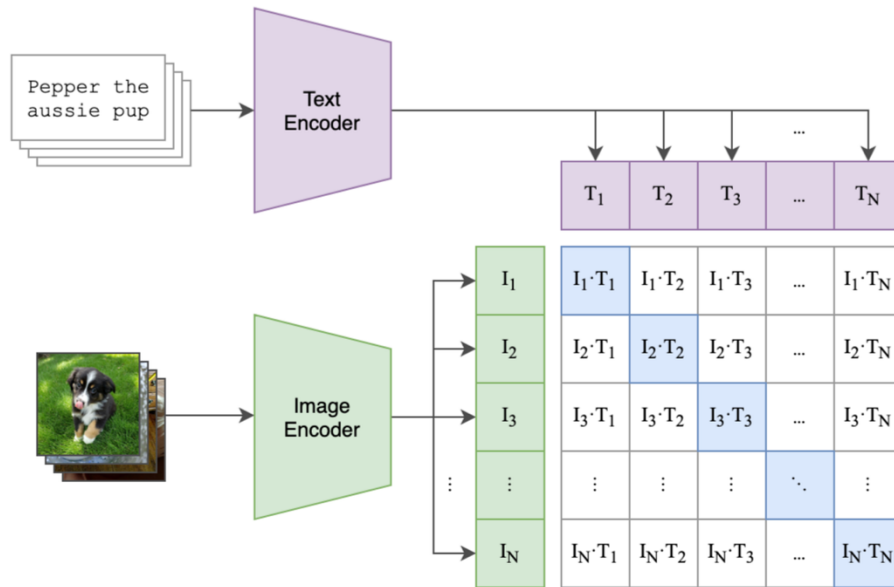
You will learn and implement a generative model in Assignment 3 that generates emojis from text inputs



Beyond 2D Recognition: Vision Language Models



Yasunaga, Michihiro, et al. "Retrieval-augmented multimodal language modeling." arXiv preprint arXiv:2211.12561 (2022).

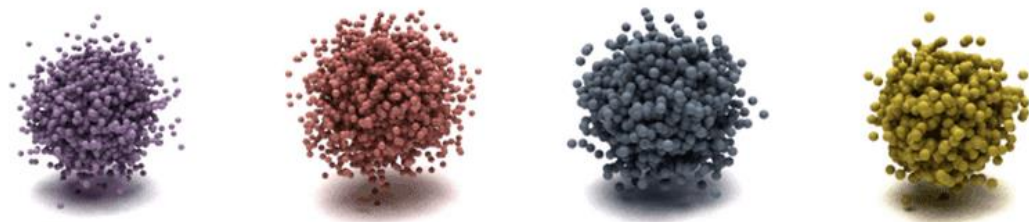


Contrastive pre-training in CLIP. The blue squares are the pairs for which we want to optimize the similarity. Image derived from <https://github.com/openai/CLIP>

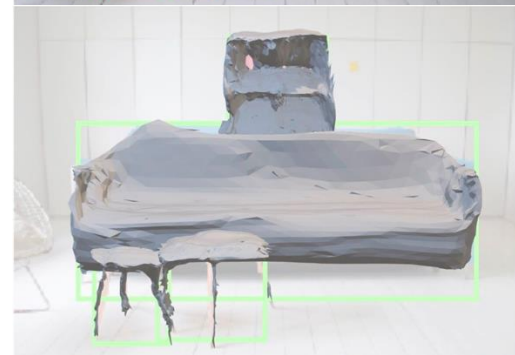
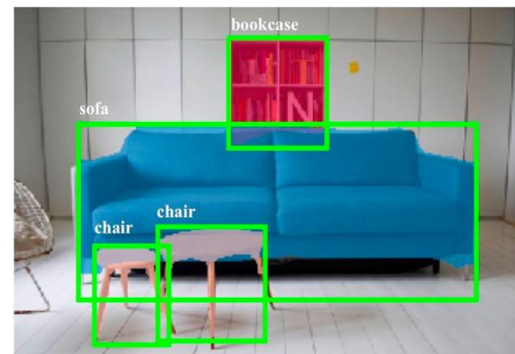
Beyond 2D Recognition: 3D Vision



Choy et al., 3D-R2N2: Recurrent Reconstruction Neural Network (2016)



Zhou et al., 3D Shape Generation and Completion through Point-Voxel Diffusion (2021)



Gkioxari et al., "Mesh R-CNN", ICCV 2019

Beyond 2D Recognition: Embodied Intelligence



Li et al., BEHAVIOR-1K: A Benchmark for Embodied AI with 1,000 Everyday Activities and Realistic Simulation (2022)



Mandlekar and Xu et al., Learning to Generalize Across Long-Horizon Tasks from Human Demonstrations (2020)

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- **Generative and Interactive Visual Intelligence**
- Human-Centered Applications and Implications

CS231n overview

- Deep Learning Basics
- Perceiving and Understanding the Visual World
- Generative and Interactive Visual Intelligence
- Human-Centered Applications and Implications

2018 Turing Award for deep learning

most prestigious technical award, is given for major contributions of lasting importance to computing.



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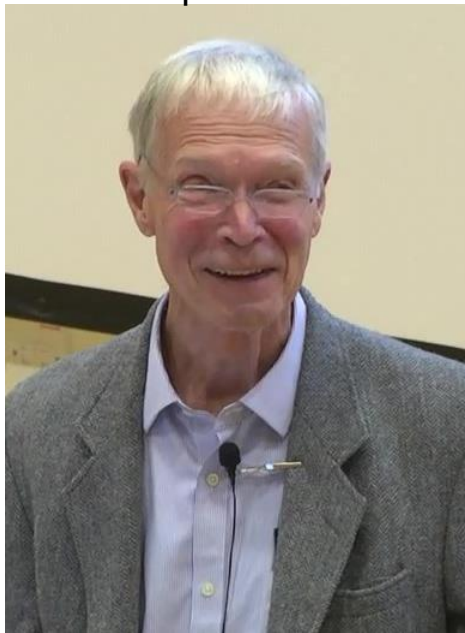
2024 Nobel Prize in Physics

Hinton speaking at the Nobel Prize
Lectures in Stockholm in 2024



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John Hopfield in 2016



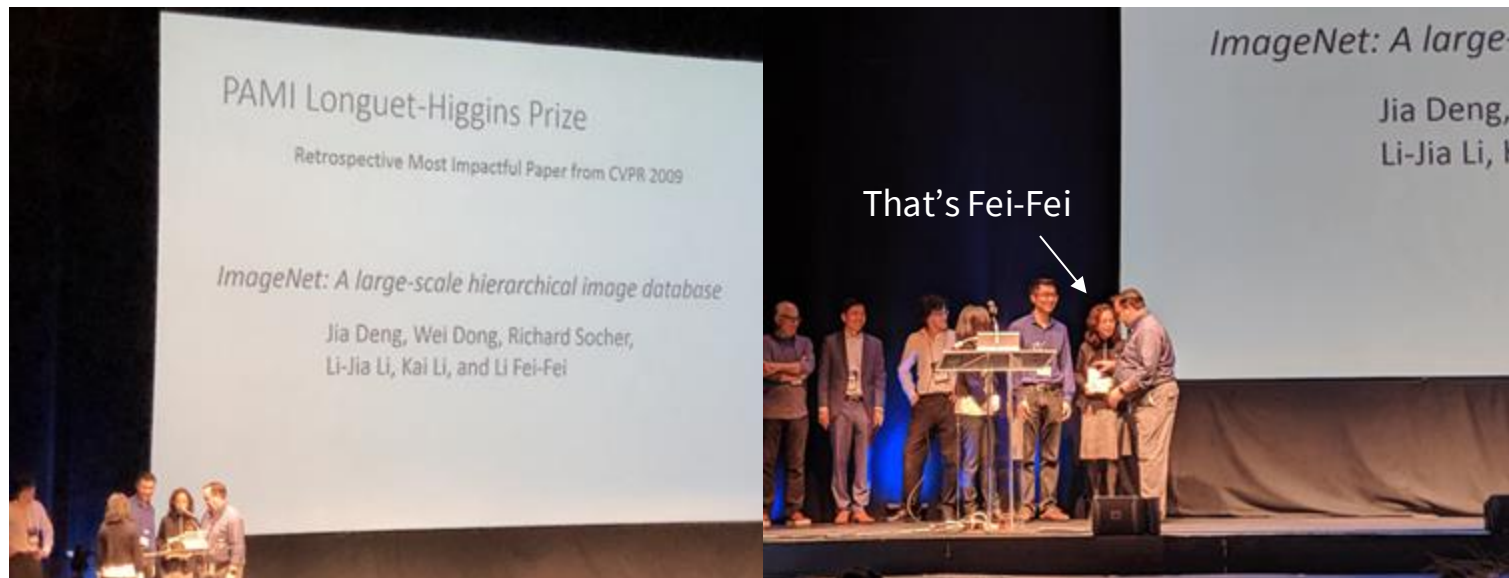
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In 2024, he was jointly awarded the [Nobel Prize in Physics](#) with [John Hopfield](#) “for foundational discoveries and inventions that enable machine learning with artificial neural networks.”

IEEE PAMI Longuet-Higgins Prize

Award recognizes ONE Computer Vision paper from ten years ago with significant impact on computer vision research.

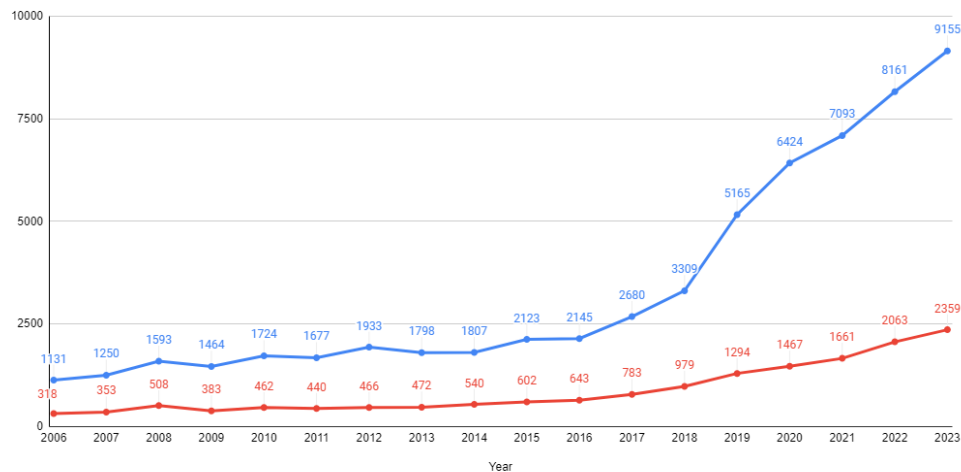
At CVPR 2019, it was awarded to the 2009 original ImageNet paper





CVPR Conference Paper Statistics

Submitted Papers Accepted Papers



Logistics

Lectures

- Tuesdays and Thursdays between 12:00 PM to 1:20 PM at NVIDIA Auditorium
- Lectures will not be streamed on Zoom but will be broadcast live via Panopto
- Slides will be posted on the course website shortly before each lecture
- All lectures will be recorded and uploaded to [Canvas](#) after the lecture under the “Panopto Course Videos” Tab.

Course website [<http://cs231n.stanford.edu/>] - Refresh!

Schedule

- **Lectures** will occur Tuesday/Thursday from 12:00-1:20pm Pacific Time at [NVIDIA Auditorium](#).
- **Discussion** sections will (generally) occur on Fridays from 12:30-1:20pm Pacific Time at [NVIDIA Auditorium](#). Check [Ed](#) for any exceptions.

Updated lecture slides will be posted here shortly before each lecture. For ease of reading, we have color-coded the lecture category titles in [blue](#), discussion sections (and final project poster session) in [yellow](#), and the midterm exam in [red](#). Note that the schedule is subject to change as the quarter progresses.

Date	Description	Course Materials	Events	Deadlines
04/01	Lecture 1: Introduction Computer vision overview Course overview Course logistics			
Deep Learning Basics				
04/03	Lecture 2: Image Classification with Linear Classifiers The data-driven approach K-nearest neighbor Linear Classifiers Algebraic / Visual / Geometric viewpoints Softmax loss			
04/04	Python / Numpy Review Session	🕒 12:30-1:20pm PT		
04/08	Lecture 3: Regularization and Optimization Regularization Stochastic Gradient Descent Momentum, AdaGrad, Adam Learning rate schedules			
04/09			Assignment 1 out	
04/10	Lecture 4: Neural Networks and Backpropagation Multi-layer Perceptron Backpropagation			
04/11	Backprop Review Session	🕒 12:30-1:20pm PT		

Friday Discussion Sections

6 Discussion sections Fridays 12:30-1:20 pm, NVIDIA Auditorium

04/04	Python / Numpy Review Session
04/11	Backprop Review Session
04/18	Final Project Overview and Guidelines
04/25	PyTorch / TensorFlow Review Session
05/02	RNNs & Transformers
05/09	Midterm Review Session

Hands-on tutorials, with more practical details than the main lecture

Check Canvas for the Zoom link for the discussion sessions! Recordings will be available on Canvas.

This Friday: Python / numpy / Colab

Ed

For questions about assignments, final project, midterm, logistics, etc, use [Ed](#)!

Access: Canvas -> Deep Learning for Computer Vision -> Ed Discussion

CGOE students: Use your @stanford.edu address to register for Ed; contact scpd-customerservice@stanford.edu for help.

Office Hours

We'll be hosting both in-person and remote office hours. (starting week 2)

- Location
 - In-person: Huang Basement, check for CS231n signs, check the course website and Canvas
 - Remote: Zoom and QueueStatus to setup queues
 - Please see [Canvas](#) or [Ed](#) for the QueueStatus link
 - TAs will admit students to their Zoom meeting rooms for 1-1 conversations when it's your turn using [QueueStatus](#).
- The office hour schedule is on the [course website](#)

Overview on communication

Course Website: <http://cs231n.stanford.edu/>

- Syllabus, lecture slides, links to assignment downloads, etc

Ed:

- Use this for most communication with course staff
- Ask questions about homework, grading, logistics, etc
- Use private questions only if your post will violate the honor code if you release it publicly

Course email address:

- cs231n-staff-spr25@stanford.edu

Gradescope:

- For turning in homework and receiving grades

Canvas:

- For watching recorded lectures
- For watching recorded discussion sessions

Assignments

All assignments will be completed using Google Colab

Assignment 1: Will be out Wednesday 4/9, due 4/23 by 11:59 PM

- K-Nearest Neighbor
- Linear classifiers: Softmax
- Two-layer neural network
- Image features
- Deep neural network and optimizers

Grading

All assignments, coding and written portions, will be submitted via [Gradescope](#).

An auto-grading system:

- A consistent grading scheme
- Public tests:
 - Students see results of public tests immediately
- Private tests
 - Generalizations of the public tests to thoroughly test your implementation

Grading

3 Assignments: 12% + 18% + 15% = 45%

In-Class Midterm Exam: 20%

Course Project: 35%

- Project Proposal: 1%
- Milestone: 2%
- Final Project Report: 29%
- Poster & Poster Session: 3%

Participation Extra Credit: up to 3%

Late policy

- 4 free late days – use up to 2 late days per assignment
- Afterward, 25% off per day late
- No late days for project report

Collaboration policy

We follow the [Stanford Honor Code](#) and the [CS Department Honor Code](#) – read them!

- Rule 1: Don't look at solutions or code that are not your own; everything you submit should be your own work
- Rule 2: Don't share your solution code with others; however discussing ideas or general strategies is fine and encouraged
- Rule 3: Indicate in your submissions anyone you worked with
- Rule 4: Do not submit AI-generated responses

Turning in something late/incomplete is better than violating the honor code

Prerequisites

Proficiency in Python

- All class assignments will be in Python (and use numpy)
- Later in the class, you will be using Pytorch and TensorFlow
- [A Python tutorial available on course website](#)

College Calculus, Linear Algebra

Optional textbook resources

- [Deep Learning](#)
 - by Goodfellow, Bengio, and Courville
 - Here is a [free version](#)
- Mathematics of deep learning
 - Chapters 5, 6 7 are useful to understand vector calculus and continuous optimization
 - [Free online version](#)
- Dive into deep learning
 - An interactive deep learning book with code, math, and discussions, based on the NumPy interface.
 - [Free online version](#)

Learning objectives

Formalize computer vision applications into tasks

- Formalize inputs and outputs for vision-related problems
- Understand what data and computational requirements you need to train a model

Develop and train vision models

- Learn to code, debug, and train convolutional neural networks.
- Learn how to use software frameworks like PyTorch and TensorFlow

Gain an understanding of where the field is and where it is headed

- What new research has come out in the last 0-5 years?
- What are open research challenges?
- What ethical and societal considerations should we consider before deployment?

Why should you take this class?

Become a vision researcher (an incomplete list of conferences)

- Get involved with [vision research at Stanford](#): apply [using this form](#).
- [CVPR 2025 conference](#)
- [ICCV 2025 conference](#)

Become a vision engineer in industry (an incomplete list of industry teams)

- [Perception team at Google AI](#), [Vision at Google Cloud](#)
- [Vision at Meta AI](#)
- [Vision at Amazon AWS](#)
- [Nvidia](#), [Apple](#), [Microsoft](#), [OpenAI](#), [Salesforce](#),

Apply computer vision to solve problems in other fields of science & engineering

General interest

CS231n: Deep Learning for Computer Vision

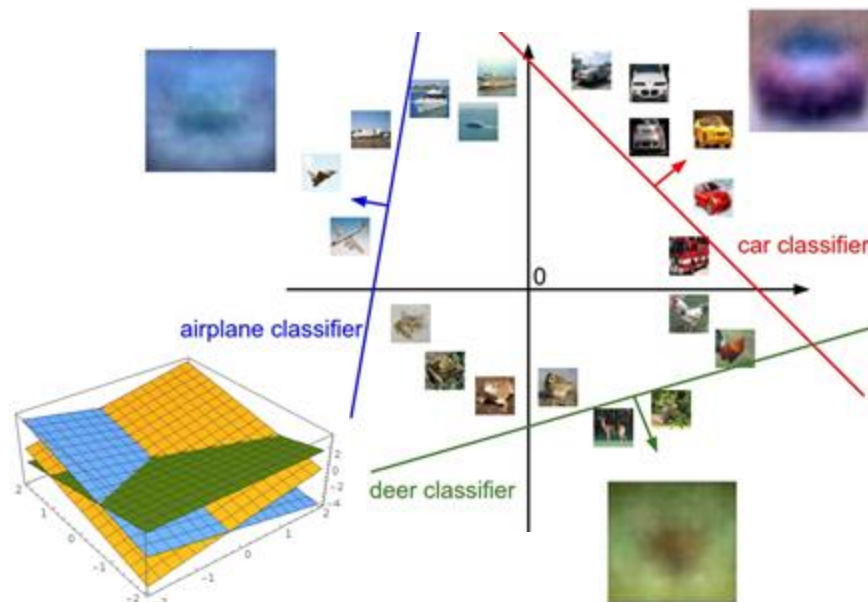
- Deep Learning Basics (Lecture 2 – 4)
- Perceiving and Understanding the Visual World (Lecture 5 – 12)
- Reconstructing and Interacting with the Visual World (Lecture 13 – 17)
- Human-Centered Artificial Intelligence (Lecture 18)

Next time: Image classification with Linear Classifiers

k- nearest neighbor



Linear classification



Plot created using [Wolfram Cloud](#)

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